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Faculty of Applied Sciences

Data Science

MASTER'S THESIS

Sergio Andres Villarreal Moyano

**Comparative Classification of Short Mental-Health-Related
Texts Using Lexical and Semantic Representations**

MASTER'S THESIS

written under the supervision of

dr inż. Godlove Kuaban

Approved

Date and supervisor's signature

Dąbrowa Górnicza 2026

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ABSTRACT

This thesis investigates the comparative classification of short mental-health-related text statements using lexical and semantic text representations. The study addresses a methodological question in mental-health-related Natural Language Processing: whether sentence embeddings provide a measurable performance advantage over strong lexical TF-IDF-based representations in a short, noisy, multiclass classification setting. The task was framed as a non-clinical supervised machine learning problem, where the objective was to classify text statements into seven predefined dataset categories rather than to provide diagnosis or clinical assessment.

A secondary labelled dataset was used for the experiment. After initial inspection, 362 empty statements were removed because they did not contain usable linguistic information, resulting in a final modelling dataset of 52,681 records. The texts were preprocessed and represented through two feature strategies: lexical TF-IDF representations and semantic sentence embeddings generated with the pretrained all-MiniLM-L6-v2 model. Seven supervised learning configurations were evaluated, including Logistic Regression, Multinomial Naive Bayes, Random Forest, and Linear SVM, depending on the representation branch. Model comparison was conducted using stratified five-fold cross-validation with Macro F1-score as the main selection metric, supported by Accuracy and Weighted F1-score. The selected model was then evaluated on a final stratified hold-out test set.

The strongest performance was obtained by the TF-IDF word-and-character Linear SVM configuration. In the final hold-out evaluation, this model achieved an Accuracy of 0.7665, a Macro F1-score of 0.7269, and a Weighted F1-score of 0.7638. The results showed that the strongest lexical configuration outperformed the tested sentence-embedding-based alternatives under the selected experimental conditions. Class-level analysis also revealed uneven performance across categories. Normal achieved the strongest result, while Stress, Suicidal, and Personality disorder remained more difficult to classify. The confusion matrix showed substantial overlap between Depression and Suicidal, suggesting that short statements assigned to these categories may share similar linguistic expressions of distress.

The findings indicate that semantic representations do not automatically outperform strong lexical baselines in short mental-health-related text classification. Instead,

representation effectiveness depends on dataset characteristics, text length, label structure, preprocessing decisions, and classifier choice. The thesis contributes a controlled and transparent comparison of lexical and semantic representation strategies while maintaining a cautious non-clinical interpretation of model outputs. The results support the need for strong lexical baselines, class-level evaluation, and responsible reporting in sensitive NLP applications.

Keywords: mental health text classification; natural language processing; TF-IDF; sentence embeddings; Linear SVM; supervised machine learning; short text classification

1. INTRODUCTION

1.1. Study Background

Mental health has become an increasingly important topic in public health, psychology, and digital research. Conditions such as depression, anxiety, stress-related disorders, and suicidal ideation continue affecting a large number of people worldwide and are associated with major personal, social and economic consequences [1], [2]. At the same time, many individuals communicate their emotional distress through written language before seeking formal support, making text a potentially valuable source of information for computational analysis.

Language can reflect emotional and cognitive tendencies through vocabulary choice, tone, repetition, and patterns of self-reference. Previous research in psychology and computational linguistics has shown that linguistic signals such as negative affect words, self-focused language, and expressions of hopelessness or worry may be associated with psychological distress [3], [4]. Although these signals should not be interpreted as clinical evidence on their own, they provide a meaningful basis for analytical research on mental-health-related text.

Recent developments in Natural Language Processing (NLP) and machine learning have expanded the range of methods available for text classification. Traditional lexical representations such as bag-of-words and TF-IDF remain widely used because they are simple, effective, and often interpretable. At the same time, semantic approaches based on transformer models and sentence embeddings have become increasingly popular because they aim to capture contextual meaning beyond surface-level word frequency [5]–[8]. In mental health research, these methods have been applied to text from social media, forums, and self-reported statements in order to detect patterns related to emotional distress or mental health conditions [9]–[12].

However, this area also raises important methodological concerns. More complex models may offer richer representations of language, but they are often harder to interpret, explain, and reproduce. In a sensitive domain such as mental health, this creates a tension between predictive performance and transparency [13], [14]. The issue becomes even more relevant when the analysed texts are short, noisy, or lexically explicit, because in

such cases it is not always clear whether semantically richer representations actually provide an advantage over simpler lexical approaches [8], [15].

For this reason, the present study focuses on the comparative classification of short mental-health-related text statements into seven predefined categories. Rather than assuming that one representation strategy is automatically superior, the study examines how lexical and semantic representations perform under a controlled supervised learning setting, while keeping methodological clarity and careful interpretation at the centre of the analysis.

1.2. Problem Statement

Despite the growing use of NLP and machine learning in mental-health-related text analysis, several methodological issues remain unresolved. One of the most important is that performance-oriented studies often give limited attention to transparency and interpretability, even though these concerns are especially important in a sensitive domain where predictions can easily be misunderstood or overinterpreted [12]–[14].

A second issue is that the relative value of lexical and semantic text representations remains unclear in short-text classification settings. Sentence embeddings are often treated as a stronger or more advanced option because they are designed to capture semantic meaning beyond surface-level word frequency. However, this assumption is not always tested under controlled conditions, and it may not hold consistently when texts are short, noisy, or lexically explicit [8], [15], [17]. In such settings, simpler lexical representations may remain highly competitive.

A third issue concerns the nature of the data itself. Many mental-health-related text datasets contain platform-dependent artefacts, informal language, lexical overlap between categories, and short statements with limited contextual information [10], [11], [19], [20]. These characteristics make classification more difficult and also complicate interpretation, because part of what a model learns may reflect dataset-specific noise as much as category-relevant linguistic signals.

Taken together, these issues create a clear methodological problem: it is not enough to compare models only in terms of predictive performance, and it is not safe to assume that semantically richer representations will necessarily outperform lexical ones in this type of task. There is a need for a controlled comparison that evaluates lexical and semantic

representations under the same supervised learning setting, while also keeping interpretability, dataset artefacts, and non-clinical framing explicitly in view.

This thesis addresses that problem through the comparative classification of short mental-health-related text statements into seven predefined categories using selected supervised learning models, with attention not only to predictive performance but also to methodological transparency and careful interpretation.

1.3. Research Purpose

The purpose of this study is to develop and evaluate a supervised machine learning framework for classifying short text statements into seven predefined mental-health-related categories using Natural Language Processing techniques.

More specifically, the study aims to compare two feature representation strategies, TF-IDF (lexical) and sentence embeddings (semantic), and to examine how these representations influence classification performance across selected supervised learning models under a controlled experimental setting. In addition to predictive performance, the study also considers methodological transparency, interpretability, and the limitations of applying automated text classification in a sensitive non-clinical domain.

This research is not intended to provide medical or clinical diagnosis. Instead, it is designed as an analytical and computational study of how linguistic patterns in short texts can be represented, classified, and interpreted in a transparent and ethically responsible way.

1.4. Goals and Research Questions

General Goal

The general goal of this study is to develop and evaluate a supervised text classification framework capable of assigning short mental-health-related statements to seven predefined categories using alternative NLP-based text representations under a controlled experimental setting.

Specific Objectives

1. To preprocess and prepare a labelled dataset of short mental-health-related text statements for supervised classification.

2. To represent the text using two feature extraction approaches: TF-IDF and semantic sentence embeddings derived from a pretrained transformer model.
3. To train and evaluate selected supervised classification models using both representation strategies.
4. To compare the performance of lexical and semantic representations using accuracy, macro F1-score, and weighted F1-score.
5. To determine which evaluated model configuration performs best under the selected experimental setting.
6. To examine class-level performance through the confusion matrix in order to identify which categories are more difficult to distinguish.
7. To analyse methodological limitations related to dataset characteristics, preprocessing decisions, representation choice, and model transparency.
8. To maintain an ethically responsible framing by treating the system as a non-clinical analytical tool rather than a diagnostic instrument.

Research Questions

1. To what extent can supervised machine learning models classify short text statements into seven predefined mental-health-related categories under the selected experimental setting?
2. Does the use of sentence embeddings improve classification performance compared with TF-IDF in this study?
3. Which evaluated model configuration performs best under the selected supervised learning setting?
4. Which categories are more difficult to distinguish, and what does the confusion matrix reveal about class-level overlap?
5. What methodological limitations emerge from the dataset, preprocessing pipeline, and model design, and how do these limitations affect the interpretation of results?

1.5. Study Scope

This study is limited to the computational classification of short text statements into seven predefined mental-health-related categories contained in the selected labelled dataset. The analysis focuses exclusively on the linguistic content of the statements and does not include demographic, behavioural, clinical, or longitudinal variables.

The methodological scope is restricted to two text representation strategies, TF-IDF and sentence embeddings, and to a selected set of supervised learning configurations evaluated under a controlled experimental setting. The study does not attempt an exhaustive benchmark of all possible classifiers, embedding models, or deep learning architectures. It also does not include transformer fine-tuning, multimodal inputs, or deployment-oriented system development.

The evaluation scope is likewise limited. Candidate configurations were compared through stratified cross-validation, and the final winning configuration was then examined through a stratified hold-out evaluation. The findings should therefore be interpreted as results obtained under the selected experimental design rather than as universal conclusions about lexical or semantic representations in all mental health text classification settings.

Finally, the study remains explicitly non-clinical in scope. The output categories are treated as dataset-defined analytical classes rather than medically validated diagnoses. For this reason, the results should be interpreted as computational classification outcomes, not as evidence of psychological assessment, screening, or clinical decision-making.

1.6. Research Significance

This study is significant for both methodological and applied reasons. From a methodological perspective, it contributes to mental-health-related NLP research by providing a controlled comparison between lexical and semantic text representations under the same supervised learning setting. This is important because the literature often assumes that semantically richer representations will necessarily outperform simpler lexical methods, even though such assumptions may not hold consistently in short and noisy text classification tasks [8], [15], [17].

The study is also significant because it does not treat model complexity as a value in itself. In a sensitive domain such as mental-health-related text analysis, stronger predictive performance is not the only issue. Transparency, reproducibility, and careful interpretation are also important, especially when the analysed texts are short, ambiguous, or affected by dataset-specific artefacts [12]–[14]. By comparing multiple selected configurations under a controlled design, the study helps clarify how representation choice and classifier behaviour interact in practice.

From an analytical perspective, the thesis is valuable because it shows that more complex semantic representations do not automatically outperform stronger lexical alternatives. In this study, the best-performing configuration came from the lexical branch rather than the embedding-based branch, which makes the comparison informative beyond a simple performance ranking. This finding supports a more cautious and evidence-based view of model selection in short-text mental-health-related classification.

The significance of this study is also connected to its clearly defined scope. The thesis focuses on the comparative behaviour of lexical and semantic representations in a short-text multiclass classification task, rather than on the development of a clinical diagnostic tool. This distinction is important because mental-health-related language is sensitive, context-dependent, and potentially ambiguous. For that reason, the contribution of the study lies not in making medical claims, but in showing how different computational representations behave when applied to the same labelled dataset under a controlled supervised learning design.

The study also covers the methodological consequences of model selection in this domain. By examining both global metrics and class-level behaviour, the thesis provides insight into whether strong general performance is evenly distributed across all categories or whether some labels remain difficult to distinguish. This is relevant because a model that performs well on average may still show important weaknesses for minority, ambiguous, or overlapping categories. Therefore, the value of the research is not limited to identifying the best-performing model, but also includes explaining what the results reveal about representation choice, dataset limitations, and responsible interpretation in mental-health-related NLP.

From a practical perspective, the study provides a reproducible framework that can support future work on mental-health-related text analysis, especially in contexts where interpretability, computational feasibility, and methodological transparency matter. Although the thesis does not propose a clinical system, it offers a structured basis for future computational research and for more critical discussion of when simpler methods may be preferable to more complex ones.

1.7. Structure of the Thesis

This thesis is organized into six chapters. Chapter 1 introduces the background, research problem, purpose, research questions, scope, and significance of the study. Chapter 2

reviews the literature on mental-health-related text classification, lexical and semantic text representations, short-text classification, and methodological concerns in sensitive NLP applications. It also presents the conceptual framework and research gap.

Chapter 3 describes the methodology, including the research approach, experimental design, dataset preparation, tools, data analysis techniques, ethical considerations, and credibility measures. Chapter 4 presents the experimental findings, including cross-validation results, final hold-out evaluation, class-level performance, confusion matrix analysis, unexpected results, and answers to the research questions.

Chapter 5 discusses the meaning of the findings in relation to previous literature, methodological limitations, and practical implications. Chapter 6 concludes the thesis by summarizing the main results, connecting them to the literature, outlining future research directions, and presenting methodological recommendations.

2. LITERATURE REVIEW

2.1. Theoretical Foundation

This study is grounded in three related theoretical and methodological foundations. First, written language can reflect aspects of emotional and psychological expression. Second, Natural Language Processing makes it possible to convert text into structured representations for computational analysis. Third, supervised machine learning can learn statistical associations between those representations and predefined categories.

The first foundation comes from research in psychology and computational linguistics showing that language use is not neutral. Word choice, emotional tone, repetition, and patterns of self-reference may reflect underlying cognitive and affective tendencies. Earlier work using tools such as LIWC showed that features such as negative emotion words, first-person singular pronouns, and absolutist expressions may be associated with distress, depression, anxiety-related expression, or other forms of psychological strain [3], [4], [16]. Although such signals should not be treated as clinical evidence on their own, they support the idea that written language contains measurable patterns that can be analysed computationally.

The second foundation concerns text representation. In NLP, the same text can be represented in different ways depending on the analytical objective. Traditional lexical approaches such as bag-of-words and TF-IDF represent language through observable terms and their statistical importance within a corpus [5], [15]. These methods are especially useful when individual words or short expressions carry strong discriminative value. In contrast, sentence embedding methods aim to encode text into dense semantic vectors that capture broader similarity in meaning and contextual information [8], [17]. These two strategies reflect different assumptions: lexical approaches emphasize explicit textual cues, while semantic approaches aim to preserve more general meaning relations beyond surface form.

The third foundation is supervised machine learning. In supervised classification, models are trained on labelled examples in order to learn associations between input features and target classes. In text classification, this means learning which feature patterns are statistically linked to particular labels. Classical models such as Logistic Regression, Naive Bayes, Random Forest, and Support Vector Machines have long been used for this purpose because they offer different trade-offs in terms of linearity, flexibility,

interpretability, and computational cost [5], [6], [18]. In particular, linear methods have often proved strong in sparse high-dimensional text settings, especially when lexical cues are clear and repeatedly associated with particular classes.

Taken together, these foundations support the logic of the present study. The thesis assumes that short mental-health-related texts may contain meaningful linguistic signals, that those signals can be represented either lexically or semantically, and that supervised models can learn useful distinctions between dataset-defined categories. At the same time, the study is guided by methodological transparency, interpretability, and explicit non-clinical framing, since in a sensitive domain such as mental-health-related text analysis predictive performance alone is not sufficient for responsible interpretation [13], [14]. This means that the study does not seek only to compare model performance, but also to do so in a way that is transparent, reproducible, and cautious in its claims.

2.2. Lexical and Semantic Text Representations

Text classification requires written language to be transformed into numerical features that can be processed by machine learning algorithms. This transformation is not a neutral technical step, because different representation strategies preserve different aspects of the original text. In the context of this thesis, the distinction between lexical and semantic representations is central because the study compares whether surface-level textual cues or denser meaning-based vectors are more effective for classifying short mental-health-related statements.

Lexical representations describe text through observable units such as words, terms, or character sequences. In traditional text classification, approaches such as bag-of-words and TF-IDF represent a document according to the presence, frequency, or relative importance of the terms that appear in it [5]. TF-IDF stands for Term Frequency-Inverse Document Frequency. It assigns weight to a term by considering both how often the term appears in a document and how rare or common that term is across the full corpus. In simplified form, it can be expressed as:

$$\text{TF-IDF}(t, d) = \text{TF}(t, d) \times \text{IDF}(t)$$

Where $\text{TF}(t, d)$ represents the frequency of term t in document d , and $\text{IDF}(t)$ reduces the weight of terms that appear in many documents while increasing the weight of more distinctive terms. As a result, words that are frequent in a specific statement but less common across the full corpus can receive greater importance. This logic is particularly

relevant in text categorization, where documents are often represented in high-dimensional sparse spaces and many individual terms may contain useful discriminative information [6].

Fig. 1 presents the two representation paths compared in this study, showing how the same short-text input can be transformed either into sparse lexical features or into a dense semantic vector.

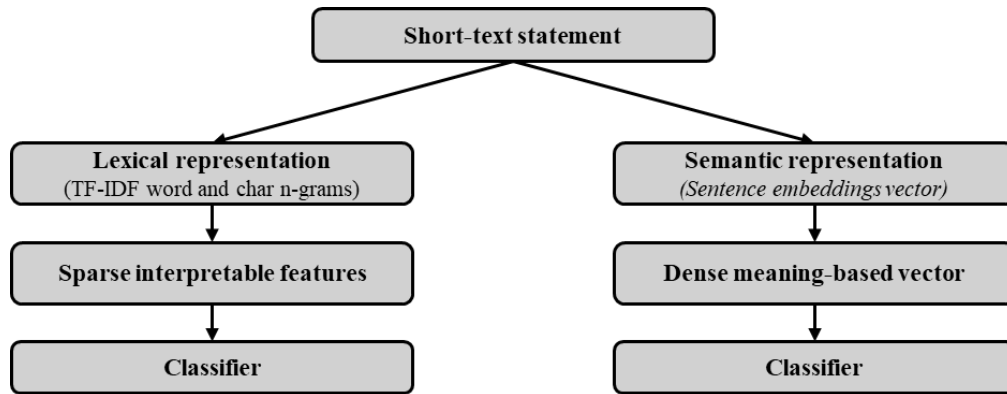


Fig. 1. Conceptual difference between lexical and semantic text representation approaches used in this study.

In short-text classification, lexical representations can be particularly effective when the target categories are associated with explicit wording. For example, expressions of worry, hopelessness, stress, fear, or self-reference may appear directly in the text and may therefore be captured through word-level features. In this setting, a lexical model does not need to infer broad meaning from a long passage; it can often rely on compact but highly informative textual signals. This is one reason why traditional lexical methods remain relevant even when more recent semantic models are available.

Another advantage of lexical representation is interpretability. Because lexical features correspond to words or character patterns, it is easier to inspect which textual cues contribute to classification decisions. This does not make the model clinically explainable in a medical sense, but it does make the computational process more transparent than a dense vector representation where individual dimensions are not directly readable. In a sensitive domain such as mental-health-related text analysis, this transparency matters because predictions can easily be overinterpreted if the modelling process is not clearly explained.

Character-level n-grams extend the lexical approach by representing text through short sequences of characters rather than full words only. This can be useful in noisy user-generated text, where spelling variation, informal expressions, punctuation, repeated letters, and short fragments may carry relevant information. A word-level model may miss some of these patterns if the exact word form is rare, while character n-grams can capture partial forms and subword regularities. For this reason, the present thesis includes a combined word-level and character-level TF-IDF representation in the lexical branch.

Semantic representations follow a different logic. Instead of representing text mainly through observable term frequencies, they encode a sentence or document as a dense numerical vector intended to capture broader meaning. Sentence embeddings are designed so that texts with similar meanings can be located closer to each other in vector space, even when they do not use the same exact words [8], [17]. This makes them attractive for mental-health-related text classification, where different individuals may express similar emotional states using different wording.

The theoretical advantage of semantic representations is that they may capture meaning beyond surface-level lexical overlap. For example, two short statements may express anxiety or distress using different vocabulary, and a sentence embedding model may represent them as semantically related even if they share few exact terms. This is the main reason why embedding-based models are often considered a stronger or more modern alternative to traditional lexical approaches.

However, this advantage should not be assumed automatically. Dense semantic representations can lose some fine-grained lexical details that are useful for classification, especially when texts are short and the available context is limited. In short statements, there is less contextual material available for the embedding model to encode, and category-relevant meaning may be concentrated in a small number of explicit words, punctuation patterns, or short expressions. In addition, pretrained sentence embedding models are usually trained on broad language data rather than on the specific label structure of a given mental-health-related dataset. Therefore, their effectiveness depends on the compatibility between the pretrained representation, the dataset, the target categories, and the classifier used on top of the embeddings.

This distinction is important for the present thesis because the study does not compare “simple” and “advanced” methods in an abstract sense. It compares two different

assumptions about how short mental-health-related statements should be represented. The lexical branch assumes that explicit words and character-level patterns contain enough discriminative signal to support classification. The semantic branch assumes that broader sentence-level meaning may provide a more useful representation than surface-level textual features. The purpose of the comparison is therefore not to confirm that one approach is universally superior, but to evaluate how both representation strategies behave under the same supervised learning design.

Previous research also supports the need for caution when comparing these approaches. Although semantic representations have become increasingly common in NLP, lexical approaches such as TF-IDF combined with linear classifiers can remain highly competitive in high-dimensional text classification settings [6]. In health-related text classification, Dessì et al. found that a TF-IDF and SVM configuration outperformed a more complex embedding-based alternative in their experimental setting [15]. This does not prove that lexical methods are generally superior, but it shows that representation complexity alone does not guarantee stronger performance.

For these reasons, the thesis treats lexical and semantic representations as complementary methodological alternatives rather than as a hierarchy from outdated to modern. Lexical representations offer transparency and direct access to explicit textual cues, while semantic representations offer a denser approximation of sentence-level meaning. In the context of this study, the comparison also helps examine whether the dataset contains discriminative lexical cues that are sufficiently informative for classification, or whether broader semantic encoding provides a measurable advantage. This makes the comparison relevant not only as a performance benchmark, but also as an investigation of how short mental-health-related statements behave as machine learning inputs.

2.3. Short Text Classification in Mental-Health-Related NLP

Short text classification is a specific challenge within Natural Language Processing because the available linguistic context is limited. Unlike longer documents, short statements often contain only a small number of words, fragments, or expressions. This makes classification more difficult because the model has less contextual information from which to infer meaning, topic, or emotional orientation. In mental-health-related NLP, this limitation is especially important because psychological distress may be expressed indirectly, briefly, or through informal language.

Mental-health-related text classification has often been studied using user-generated content from platforms such as Twitter, Reddit, forums, and other online environments [9]–[11], [19], [20]. These sources are valuable because individuals may express emotional states, distress, worry, or suicidal ideation in written form before seeking formal support. However, this type of data is also methodologically difficult. User-generated texts may include spelling variation, abbreviations, punctuation patterns, emojis, repeated characters, platform-specific expressions, and incomplete sentences. These characteristics can introduce noise into the dataset and make classification less stable.

A further challenge is that mental-health-related categories are not always linguistically separate. Some categories may share similar words or emotional expressions, even if they represent different labels in the dataset. For example, texts related to depression, stress, anxiety, or suicidal ideation may all contain negative affect, self-reference, hopelessness, worry, or expressions of exhaustion. As a result, classification errors may occur not only because the model is weak, but also because the categories themselves overlap at the language level. This is one reason why class-level analysis is necessary in addition to global performance metrics.

The problem becomes more complex in multiclass classification. In binary classification, the model only needs to distinguish between two categories, such as depressed and non-depressed, or risk and no risk. In the present study, however, each statement must be assigned to one of seven predefined categories. This increases the difficulty of the task because the model must learn boundaries between several related classes rather than separating only one target condition from a control group. Therefore, strong global performance does not necessarily mean that all categories are classified equally well.

This issue is particularly relevant in mental-health-related NLP because prediction outputs can be easily misunderstood. The labels used in this thesis are treated as dataset-defined categories, not as clinical diagnoses. A model may learn statistical associations between language patterns and labels, but it does not assess a person’s mental health condition in a medical sense. Previous literature has emphasized that mental-health prediction from social media and other digital traces requires careful interpretation, transparent reporting, and awareness of ethical risks [12]–[14]. For this reason, the

present study avoids diagnostic claims and frames the task as computational text classification.

Evaluation metrics must also be selected carefully. Accuracy measures the proportion of correctly classified instances over the total number of instances, but it can be misleading in multiclass and imbalanced settings because it may hide poor performance on smaller or more difficult categories. A model can achieve acceptable accuracy while still misclassifying minority classes or categories with overlapping language. For this reason, the present thesis uses Macro F1 as the main comparison metric, supported by Accuracy and Weighted F1.

The F1-score combines precision and recall into a single measure. For each class, precision indicates how many of the instances predicted as that class were correct, while recall indicates how many of the actual instances of that class were correctly identified. The class-level F1-score is calculated as:

$$F1 = 2 \times \frac{(Precision \times Recall)}{(Precision + Recall)}$$

The factor 2 appears because the F1-score is calculated as the harmonic mean of precision and recall. It balances both measures in a single value and penalizes models that perform well on one of them but poorly on the other. Macro F1 is then obtained by calculating the F1-score separately for each class and averaging the class-level scores equally across all classes. In this study, this is important because the task involves seven categories and the objective is not only to perform well on the most frequent labels, but also to evaluate whether the model performs reasonably across the full label set. Weighted F1 is also reported because it accounts for class frequency, while Accuracy is included as a general performance indicator.

The confusion matrix is also important because it shows where the model makes specific errors. In mental-health-related short text classification, this is more informative than reporting only a single score. If two categories are frequently confused, this may indicate lexical overlap, ambiguous expressions, insufficient contextual information, class imbalance, or limitations in the representation strategy. Therefore, the confusion matrix supports a more cautious interpretation of the results and helps identify which categories remain difficult to distinguish.

In this thesis, short mental-health-related text classification is therefore understood as a supervised learning problem with methodological and interpretive constraints. The aim is not only to obtain the highest possible score, but also to examine how lexical and semantic representations behave under the same task conditions. This framing is important because it connects the technical design of the study with the ethical and methodological limitations of applying machine learning to sensitive mental-health-related language.

2.4. Review of Existing Studies

Research on mental-health-related text analysis has expanded considerably over the last decade. Much of this work has focused on identifying linguistic patterns associated with depression, anxiety, suicidal ideation, stress, or related forms of emotional distress in user-generated text [9]–[12]. Although the methods and data sources differ across studies, the literature broadly supports the view that written language can contain computationally useful signals for mental-health-related analysis. At the same time, this research area also presents persistent methodological challenges related to data quality, label validity, platform-specific noise, model transparency, and responsible interpretation.

One important group of studies has examined how digital language reflects psychological states or behavioural change. De Choudhury, Counts, and Horvitz showed that social media data could reveal emotional and behavioural shifts related to postpartum depression [9]. Similarly, Coppersmith and colleagues demonstrated that mental-health-related signals could be studied computationally through large-scale Twitter data, using features such as LIWC categories, n-grams, and pattern-of-life indicators [10], [19]. These studies were important because they showed that distress-related patterns can appear in naturally occurring digital communication. However, they also exposed the field to problems such as noisy labels, self-disclosure bias, uncertain ground truth, and platform-dependent language.

A related line of research has focused on early detection tasks. Losada and Crestani introduced the eRisk shared task on early risk detection, where the objective was not only to classify users but to identify signs of depression as early as possible from sequential user-generated content [11]. This type of work is relevant because it shows that evaluation design strongly affects the interpretation of results. Early detection differs from ordinary post-level or statement-level classification because it considers the timing and accumulation of evidence. Although the present thesis does not perform early detection,

this literature is useful because it demonstrates that mental-health-related NLP results depend not only on the model, but also on the task definition and evaluation setting.

A second group of studies has relied on lexical representations and classical machine learning models. In many text classification settings, bag-of-words, TF-IDF, and n-gram-based features remain competitive because they preserve explicit lexical cues that may be strongly associated with the target labels. Pirina and Çöltekin, for example, reported strong performance for depression identification on Reddit using Linear SVM with character and word n-grams [20]. Dessì et al. also found that TF-IDF combined with SVM outperformed a more complex embedding-based alternative in a health-related text classification setting [15]. These studies are especially relevant for the present thesis because they challenge the assumption that richer semantic representations necessarily produce better classification results.

A third line of research has shifted toward semantic and neural text representations. Transformer-based approaches and sentence embeddings have become increasingly common because they aim to capture semantic similarity beyond direct word overlap [8], [17]. In mental-health-related text analysis, this shift has been driven by the expectation that contextual representations may better capture subtle, indirect, or varied expressions of distress. However, the expected advantage of semantic representations has not been uniformly confirmed across tasks and datasets. This issue is especially relevant in short-text settings, where limited context may reduce the advantage of dense semantic representations and where a small number of explicit lexical cues may carry substantial discriminative information. Their effectiveness depends on factors such as text length, domain specificity, label structure, training data, and the classifier used on top of the representation.

The literature also reveals several recurring methodological limitations. First, many studies are based on platform-specific corpora such as Twitter or Reddit, where texts may include informal spelling, repost fragments, abbreviations, topic-specific vocabulary, and other artefacts that can influence both training and interpretation [10], [11], [19], [20]. Second, mental-health-related labels are often difficult to validate because online expressions are not equivalent to clinical assessment. Third, some studies emphasize predictive performance more than interpretability, reproducibility, or careful reporting of

modelling decisions [12]–[14]. These limitations are important because they affect how far computational results can be generalized beyond the dataset used in a particular study.

Another important issue is that many studies differ in several dimensions at the same time: data source, label definition, preprocessing strategy, feature representation, classifier, evaluation split, and performance metric. This makes direct comparison difficult. A model may appear stronger because of the representation used, but the observed difference may also come from the dataset, class distribution, validation design, or reporting practice. For this reason, controlled comparisons are necessary when the purpose is to examine the effect of representation choice. The present thesis responds to this issue by comparing lexical and semantic representations under the same dataset, the same general preprocessing logic, and the same evaluation framework.

Table I summarizes selected studies that are methodologically relevant to the present thesis. The table does not aim to provide an exhaustive review of all mental-health-related NLP research. Instead, it highlights studies that help justify the comparison between lexical and semantic representations, the use of supervised classification, and the need for careful interpretation of model results.

Table I. Summary of selected related work and methodological relevance to this thesis

Study	Data / Task	Methods	Key Result	Main Limitation	Relevance to this thesis
De Choudhury et al. (2013) [9]	Twitter; 376 mothers. Task: predicting postpartum changes	SVM with behavioural and linguistic features	Performance improved from prenatal prediction to early postnatal detection	Proxy outcomes, not clinical diagnosis	Shows that digital language and behaviour can reflect mental-health-related change
Coppersmith et al. (2014) [10]	Twitter; bipolar, depression, PTSD, SAD. Task: quantifying mental-health-related signals	LIWC, n-grams, pattern-of-life features, linear classifier	Predictive signal identified across multiple conditions	Self-disclosed diagnoses and noisy controls	Supports linguistic prediction, but highlights platform noise and label uncertainty

Losada & Crestani (2016) [11]	Reddit. Task: early depression detection	ERDS, Logistic Regression, dynamic prediction stream	Early depression detection was achievable, but performance depended on the evaluation setup	Early detection setting differs from standard one-shot classification	Shows that task design and evaluation setting affect result interpretation
Chancellor & De Choudhury (2020) [12]	Review of 75 papers. Task: methodological review of mental health NLP	Systematic review and critical analysis	Only 42% of reviewed studies met reporting standards	Review paper, not an empirical benchmark	Supports methodological transparency and caution in model reporting
Dessi et al. (2020) [15]	n2c2 clinical notes. Task: morbidity identification	TF-IDF + SVM vs. BiLSTM + embeddings	TF-IDF + SVM outperformed the embedding-based approach	Clinical notes differ from social-media text	Suggests that lexical models can outperform more complex semantic ones
Pirina & Çöltekin (2018) [20]	Reddit. Task: depression identification	Linear SVM with word and character n-grams	Strong in-domain performance, but weaker cross-topic generalization	Performance drops under topic shift	Supports the relevance of lexical modelling for short and noisy text

Taken together, the studies summarized in Table I show that mental-health-related text classification is feasible, but also methodologically sensitive. Prior work supports the use of linguistic and computational signals, while also showing that results depend strongly on data source, label definition, representation strategy, model choice, and evaluation design. This is directly relevant to the present thesis, which compares lexical and semantic representations under the same supervised learning framework rather than assuming that one representation type is automatically superior.

In short, the reviewed studies justify the relevance of the present research while also clarifying its position in the field. Existing work shows that mental-health-related text classification is possible, but it also leaves open an important methodological question: whether semantically richer representations actually outperform strong lexical

alternatives in short-text classification under a controlled supervised setting. The present thesis addresses this focused methodological problem by comparing selected lexical and semantic configurations on the same dataset, under the same evaluation framework, and with explicit attention to transparency, class-level behaviour, and methodological limitations.

2.5. Conceptual Framework

The conceptual framework of this study illustrates how short mental-health-related text statements are transformed into alternative computational representations and evaluated under a controlled supervised learning setting. Rather than treating text classification as a single-model task, the framework is designed to compare multiple candidate configurations in order to examine how different representation strategies influence classification performance.

The starting point of the framework is a collection of short written user statements assigned to seven predefined categories. These statements are treated as observable linguistic inputs from which computationally useful patterns may be extracted. Although the texts are not interpreted as clinical evidence, they are assumed to contain lexical and semantic signals that can be associated with the labelled categories through supervised learning [3], [4].

The next stage is text processing. At this stage, the raw text is cleaned and normalized in order to reduce noise and prepare the data for feature extraction. This step is particularly important in user-generated text, where inconsistent spelling, informal wording, and source-specific artefacts are common.

After preprocessing, the framework branches into two representation paths. The first path is lexical representation through TF-IDF features, which capture the explicit presence and relative importance of words and subword patterns in the text [5]. The second path is semantic representation through sentence embeddings, where each statement is encoded as a dense numerical vector intended to preserve broader contextual meaning beyond direct word frequency [8], [17].

These two representation strategies are then combined with supervised classifiers to form seven candidate model configurations. In the lexical branch, the tested models are Logistic Regression, Naive Bayes, Random Forest, and Linear SVM with a combined word-and-character TF-IDF representation. In the embedding branch, the tested models

are Logistic Regression, Random Forest, and Linear SVM. This design allows the study to compare lexical and semantic representations under the same dataset and general evaluation conditions, while also examining how classifier choice interacts with representation type.

The comparison stage is based on stratified five-fold cross-validation, which is used to estimate model performance more reliably and reduce dependence on a single train-test split. Because the study is multiclass and the category distribution is not perfectly balanced, model comparison is centred primarily on Macro F1, supported by Accuracy and Weighted F1. After cross-validation, the best-performing configuration is selected and evaluated on a final hold-out test set.

The final stage of the framework is evaluation and interpretation. At this stage, the selected model is assessed using final performance metrics, a classification report, and a confusion matrix in order to identify both overall performance and class-level confusion. In this way, the framework supports not only a comparison of predictive results, but also a transparent interpretation of where the model performs well and where misclassification remains [13], [14].

Figure 2 presents the conceptual workflow followed in this thesis, from the initial short user statements to the final model evaluation stage.

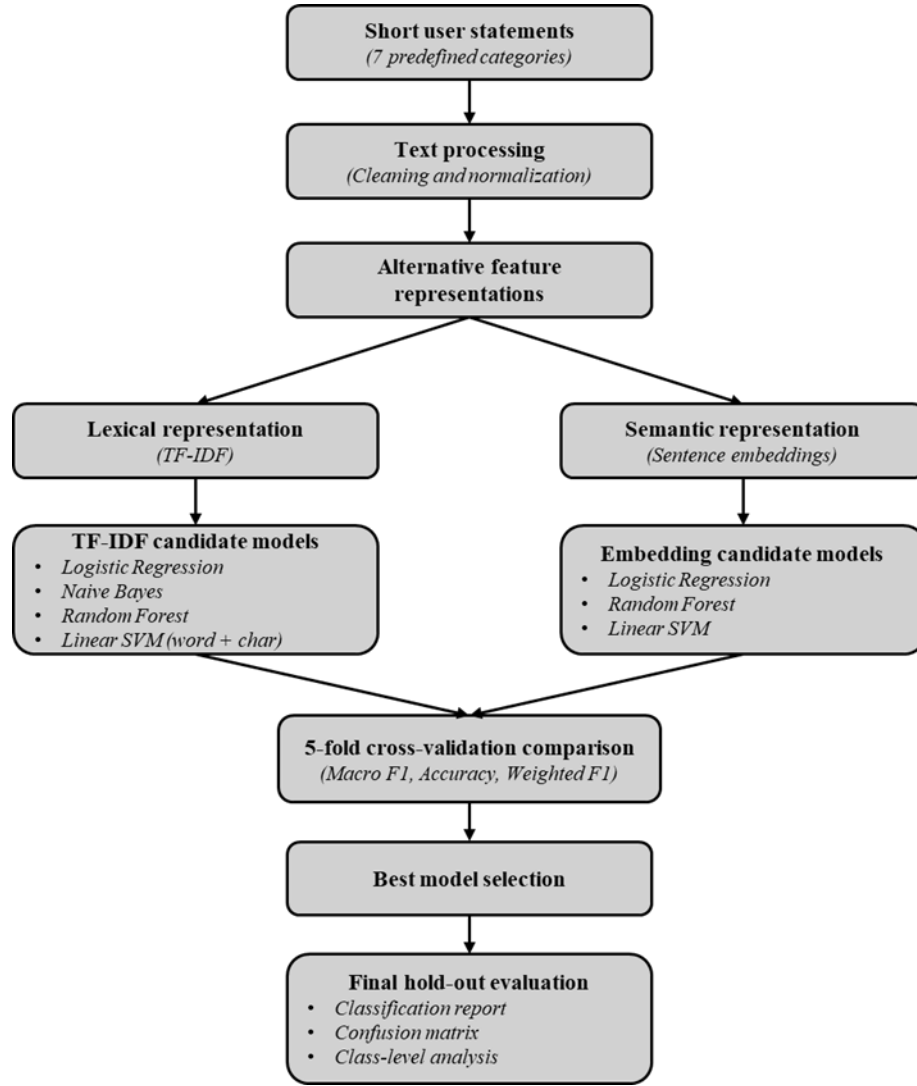


Fig. 2. Conceptual framework of the study.

As shown in Fig. 2, the framework begins with short user statements assigned to seven predefined categories. These statements are processed and then represented through two alternative feature paths: lexical TF-IDF features and semantic sentence embeddings. Each representation branch is combined with selected supervised learning classifiers, producing seven candidate configurations. The configurations are compared through stratified five-fold cross-validation using Macro F1, Accuracy, and Weighted F1. The best-performing configuration is then evaluated on a final hold-out test set using a classification report, a confusion matrix, and class-level analysis.

In summary, the conceptual framework reflects the core logic of the study: short user texts are preprocessed, represented through either lexical or semantic features, classified through multiple supervised learning configurations, compared using cross-validation,

and finally evaluated through both overall and class-level performance analysis in a controlled, non-clinical setting.

2.6. Research Gap

Although previous research has shown that mental-health-related text classification is feasible, several important gaps remain. First, much of the existing literature focuses on detecting a single condition, most commonly depression, or on binary classification settings rather than on broader multiclass categorization problems involving several related mental-health-related labels [10], [11], [20], [21]. This limits the extent to which prior findings can be generalized to short-text classification tasks where multiple related categories must be distinguished within the same dataset.

Second, the literature still provides limited controlled comparison between lexical and semantic representation strategies under comparable experimental conditions. Many studies evaluate only one representation family, use different datasets, or vary feature representation, classifier choice, validation design, and performance metrics at the same time [12], [15], [22]. As a result, it is often difficult to determine whether observed performance differences are mainly caused by the representation strategy itself, the classifier, the dataset, or the evaluation procedure.

Third, the relative value of semantic representations remains especially unclear in short-text settings. Transformer-based embeddings and sentence-level semantic vectors are often expected to capture richer meaning than lexical methods, but short statements provide limited contextual information. In such cases, explicit lexical cues, word patterns, character n-grams, punctuation, or short expressions may carry strong discriminative information. Therefore, it cannot be assumed that semantic representations will always outperform strong lexical baselines in short mental-health-related text classification.

Fourth, although recent work increasingly emphasizes explainability, transparency, and careful methodological reporting, these aspects are still not consistently integrated into comparative machine learning studies in this domain [12]–[14], [21], [22]. In mental-health-related text analysis, this is especially important because model performance alone is not sufficient. The evaluation must also clarify how results were obtained, how individual classes behave, where misclassification remains, and why the outputs should not be interpreted as clinical diagnosis.

Finally, while the field is evolving toward more complex settings such as timeline-level modelling, evidence extraction, and summarization, this does not remove the need for well-scoped and methodologically transparent studies at the short-text classification level [23]. Without controlled comparisons at this level, it remains difficult to understand whether simpler lexical models are still competitive against semantic representations when both are evaluated under the same dataset, validation strategy, and performance metrics.

The present thesis addresses this gap by comparing selected lexical and semantic representation strategies for the supervised classification of short mental-health-related statements into seven predefined categories. The study evaluates seven candidate configurations under the same dataset and evaluation framework, using stratified five-fold cross-validation for model comparison and a final hold-out evaluation for the selected model. By combining overall metrics with class-level analysis through the confusion matrix, the thesis contributes a focused, transparent, and non-clinical comparison of representation strategies in short mental-health-related text classification.

3. METHODOLOGY

3.1. Research Approach

This study adopts a quantitative research approach. The choice of this approach is determined by the nature of the research problem, the structure of the dataset, and the analytical procedures used in the study. More specifically, the research is based on a labelled corpus of short mental-health-related text statements, which are transformed into numerical feature representations and evaluated through supervised machine learning models under a controlled experimental setting.

A quantitative approach is appropriate because the study seeks to measure and compare classification performance across alternative modelling configurations rather than to explore subjective experiences or interpret meanings through qualitative inquiry. The main objective is to determine how different text representation strategies influence predictive results in a multiclass classification task. This requires numerical analysis of model outputs through performance metrics such as Accuracy, Macro F1-score, and Weighted F1-score, as well as structured comparison through stratified cross-validation and final hold-out testing.

The study is therefore empirical and comparative in nature. It does not rely on interviews, thematic coding, or mixed-method integration. Instead, it uses computational text processing, feature extraction, supervised learning, and metric-based evaluation to generate reproducible findings. The quantitative design also supports methodological transparency, since the same dataset, preprocessing decisions, and evaluation criteria are applied across all candidate configurations.

In this sense, the study treats mental-health-related text classification as a measurable machine learning problem in which differences between lexical and semantic representations can be assessed through consistent numerical evidence. The selected approach is therefore appropriate for answering the research questions and for producing results that are interpretable, comparable, and aligned with the non-clinical scope of the thesis.

3.2. Design of the Study

This study follows a comparative experimental design within a supervised machine learning setting. The design is comparative because it systematically evaluates alternative

text representation strategies and classifier configurations under the same dataset and general evaluation protocol. It is also experimental in the sense that the study intentionally varies selected modelling components, namely feature representation and classifier type, in order to observe how these choices affect classification performance.

More specifically, the study compares two main representation branches: a lexical branch based on TF-IDF features and a semantic branch based on sentence embeddings. Within these branches, seven candidate supervised learning configurations are evaluated. This structure makes it possible to examine not only whether lexical and semantic representations differ in predictive effectiveness, but also whether their performance changes depending on the classifier paired with each representation.

The design is controlled rather than exploratory. The same labelled dataset, the same preprocessing logic, and the same evaluation strategy are applied across all candidate configurations in order to reduce unnecessary variation between experiments. This is important because the purpose of the study is not simply to identify the single highest score, but to make a fair comparison between alternative modelling strategies under comparable conditions.

The study also includes two evaluation stages. In the first stage, candidate models are compared through stratified five-fold cross-validation in order to obtain more stable estimates of performance across splits of the training data. In the second stage, the best-performing configuration is evaluated on a final stratified hold-out test set. This two-step design supports both model selection and final validation while reducing dependence on a single train-test partition.

In sum, the design of the study is aligned with the research objective of determining how representation strategy influences multiclass mental-health-related text classification. By combining controlled comparison, repeated validation, and final hold-out testing, the design supports a transparent and methodologically coherent analysis of predictive performance in a non-clinical computational setting.

Table II. Candidate model configurations evaluated in the study

Branch	Representation	Classifier
Lexical	TF-IDF	Logistic Regression
Lexical	TF-IDF	Multinomial Naive Bayes

Lexical	TF-IDF	Random Forest
Lexical	TF-IDF word + character	Linear SVM
Semantic	Sentence embeddings	Logistic Regression
Semantic	Sentence embeddings	Random Forest
Semantic	Sentence embeddings	Linear SVM

Table II summarizes the seven candidate configurations evaluated in the study. The lexical branch includes three standard TF-IDF configurations and one combined word-and-character TF-IDF configuration with Linear SVM. The semantic branch includes three models trained on sentence embeddings. This structure allows the study to compare representation strategies while also examining how classifier choice affects performance within each branch.

3.3. Dataset and Sampling Strategy

This study uses a secondary labelled dataset rather than data collected directly from human participants. Therefore, the sampling strategy does not involve participant recruitment, survey distribution, or clinical assessment. The original dataset consisted of short text statements already assigned to one of seven predefined mental-health-related categories.

The original dataset contained 53,043 records and three columns: Unnamed: 0, statement, and status. The column Unnamed: 0 was treated as an index column and removed before modelling. The statement column contained the raw textual input, while the status column contained the target category for supervised classification. After preprocessing, the cleaned text column ‘statement_clean’ was used as the model input, and status was used as the target label.

During the initial inspection, 362 empty statements were identified. These records were removed before modelling because they did not contain usable linguistic information for text classification. This reduced the dataset from 53,043 to 52,681 records. Very short but non-empty statements were retained, since short expressions may still contain meaningful lexical or emotional cues in short-text classification.

Table III. Dataset structure used in the study

Element	Description
Original dataset size	53,043 records

Empty statements removed	362 records
Final dataset size used for modelling	52,681 records
Original columns	Unnamed: 0, statement, status
Removed column	Unnamed: 0
Raw text column	Statement
Cleaned text input	statement_clean
Target column	Status
Number of categories	7
Evaluation split	Stratified 80/20 hold-out after model selection

The unit of analysis is the individual text statement. Each statement is treated as one instance for supervised classification. The target population, in methodological terms, is not a clinical population, but a collection of short user-generated texts that contain language related to mental-health concerns. For this reason, the dataset is suitable for studying computational classification behaviour, but it cannot be treated as representative of people with clinically confirmed mental health conditions.

The dataset was selected through purposive secondary data sampling. It was chosen because it contains labelled short texts, multiple mental-health-related categories, and enough structure to support a supervised multiclass classification task. This makes it appropriate for the purpose of comparing lexical and semantic text representations under a controlled experimental setting.

After empty-statement removal, the dataset remained unevenly distributed across the seven categories. The largest categories were Normal, Depression, and Suicidal, while Personality disorder remained the smallest category. This imbalance is important because it can affect model performance and may cause global metrics such as Accuracy to present an incomplete view of class-level behaviour.

Table IV. Class distribution after empty-statement removal

Category	Number of statements	Percentage
Normal	16,343	31.02%
Depression	15,404	29.24%
Suicidal	10,652	20.22%
Anxiety	3,841	7.29%
Bipolar	2,777	5.27%
Stress	2,587	4.91%

Personality disorder	1,077	2.04%
Total	52,681	100.00%

The sampling process for model evaluation was handled through stratified splitting. Stratification was used to preserve the relative distribution of the seven categories across cross-validation and final hold-out evaluation. This was necessary because the task is multiclass and the category distribution is uneven. Without stratification, smaller categories could be underrepresented in some splits, leading to less reliable performance estimates.

This sampling strategy has clear limitations. Since the study relies on an existing dataset, the researcher does not control how the original texts were created, collected, or labelled. Prior research on mental-health-related social media datasets has shown that dataset choice, labelling assumptions, platform effects, and corpus composition can strongly affect classification results and may lead to misleading conclusions if they are not reported carefully [12], [20]. For this reason, the findings of this thesis are interpreted as dataset-specific computational results rather than general conclusions about mental health status in real populations.

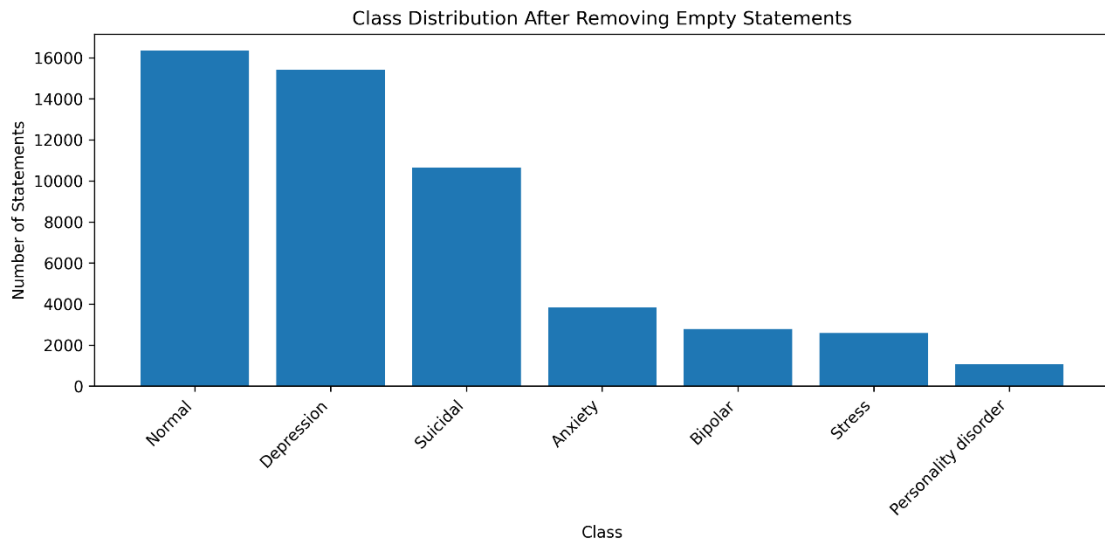


Fig. 3. Class distribution after empty-statement removal.

Fig. 3 illustrates the uneven distribution of the seven categories after empty-statement removal. Normal, Depression, and Suicidal account for most of the dataset, while Personality disorder is the smallest category. This imbalance supports the use of stratified splitting and Macro F1 as the main model-selection metric.

3.4. Tools and Instruments

The study was implemented in Python using a Google Colab notebook environment. Google Colab was used because it allows data loading, preprocessing, model training, evaluation, visualization, and export of results to be organized within a single reproducible computational workflow. The notebook-based structure also made it possible to keep the main stages of the experiment visible and traceable, from dataset inspection to final model evaluation.

The dataset was handled using pandas and NumPy. These libraries supported data loading, column inspection, class distribution analysis, removal of empty statements, construction of input and target variables, and export of tabular outputs. The main modelling input was the cleaned text column ‘statement_clean’, while the target variable was the ‘status’ column.

Matplotlib was used to generate visual outputs for the thesis, including the class distribution plot, cross-validation performance charts, final hold-out metric chart, class-level F1-score chart, and confusion matrix. The main tables and figures were exported from the notebook as CSV and PNG files, which helped preserve consistency between the computational results and the thesis document.

For text preprocessing, the study used NLTK resources, including English stopwords and WordNet-based lemmatization. The preprocessing function applied normalization steps such as lowercasing, replacement of non-selected special characters, whitespace normalization, selective stopword filtering, and lemmatization for the lexical representation branch. Negation and self-reference terms were preserved because they may carry relevant meaning in mental-health-related short texts. Empty statements were removed before modelling because they did not contain usable linguistic information.

The preprocessing strategy was intentionally conservative. Since the study worked with short statements, overly aggressive cleaning could remove potentially meaningful lexical cues. For example, a URL or platform-related fragment may contain a mental-health-related term within it, and removing the whole fragment could also remove that signal. However, this choice also means that some URL- or platform-related tokens may remain in the processed text. For this reason, these tokens were inspected later as potential artefacts and were treated as a methodological limitation rather than as clinically meaningful linguistic evidence.

The lexical representation branch was implemented with scikit-learn’s TF-IDF vectorizer. TF-IDF was selected because it represents text through weighted term-based features and remains a standard baseline in text classification. In this study, TF-IDF was applied with word-level n-grams, and one candidate configuration also combined word-level and character-level TF-IDF features. This allowed the lexical branch to capture both common word patterns and shorter subword patterns that may be useful in noisy short texts.

The semantic representation branch was implemented using the sentence-transformers library. The pretrained model all-MiniLM-L6-v2 was used to generate dense sentence embeddings for each cleaned statement. Sentence embeddings were included because they represent text as semantic vectors rather than sparse term-frequency features, allowing direct comparison between lexical and semantic representation strategies. This choice is consistent with prior work on Sentence-BERT, which introduced sentence embeddings designed to capture semantic similarity between sentences [8].

The supervised learning models were implemented using scikit-learn. Seven candidate configurations were evaluated: TF-IDF with Logistic Regression, TF-IDF with Multinomial Naive Bayes, TF-IDF with Random Forest, TF-IDF word-and-character features with Linear SVM, sentence embeddings with Logistic Regression, sentence embeddings with Random Forest, and sentence embeddings with Linear SVM. These models were selected because they reflect different learning assumptions and are widely used in supervised text classification. Linear SVM is especially relevant in text classification because previous work has shown its suitability for high-dimensional text data [6], while Random Forest provides an ensemble-based comparison model [18].

Model evaluation was also carried out with scikit-learn. The study used stratified five-fold cross-validation and a final stratified hold-out evaluation. The main evaluation outputs were Accuracy, Macro F1-score, Weighted F1-score, classification reports, and the confusion matrix. Macro F1-score was treated as the main selection criterion because the task involves seven categories and class-level balance is more informative than overall accuracy alone.

To support reproducibility and result traceability, the final implementation notebook and exported computational outputs were organized in a GitHub repository. The repository contains the implementation notebook, exported CSV files, generated PNG figures, and supplementary materials used to support the thesis results. This provided a clear

connection between the numerical results and visual outputs reported in the thesis and the computational workflow that generated them.

3.5. Data Collection Process

This study does not involve primary data collection from human participants. Instead, it relies on a pre-existing labelled dataset of short mental-health-related text statements used as secondary research material. For this reason, the data collection process in this thesis refers to dataset acquisition, loading, inspection, cleaning, and preparation for computational analysis rather than to surveys, interviews, or clinical observation.

The dataset was obtained in digital form and loaded into the Google Colab environment for structured analysis. After loading, the data were examined to confirm the presence of the required text and label fields, identify the seven target categories, inspect class distribution, and assess whether the corpus was suitable for supervised multiclass classification. This stage also included inspection of empty and very short statements before preprocessing.

During the inspection stage, 362 empty statements were identified. These records were removed before modelling because they did not contain usable linguistic information for text classification. Very short but non-empty statements were retained because short expressions may still contain meaningful lexical or emotional cues in short-text classification. After this cleaning step, the dataset used for modelling contained 52,681 records.

Because the study is based on secondary labelled text data, the researcher did not control the original process through which the texts were created, collected, or annotated. The role of data collection in this study was therefore limited to dataset acquisition and computational preparation within the selected research environment. After inspection and empty-statement removal, the text data were passed to the preprocessing stage, where the statements were normalized and prepared for lexical and semantic feature extraction.

The process was conducted within a notebook-based workflow in Google Colab, which allowed data loading, preprocessing, model comparison, final evaluation, and export of outputs to be organized in a single computational environment. This helped maintain consistency across the main stages of the experiment and supported transparent handling of the dataset during the analysis.

This process has clear limitations. Since the study uses an existing corpus, the researcher cannot verify all original collection conditions, labelling assumptions, or contextual factors surrounding the source texts. For this reason, the dataset is treated as a bounded computational resource for controlled text classification experiments rather than as direct evidence of clinical mental health status in real populations.

3.6. Data Analysis Techniques

The study applies computational and quantitative data analysis techniques to examine how different text representation strategies affect multiclass classification performance. The analysis is structured around a supervised machine learning workflow in which short text statements are first cleaned, transformed into alternative feature representations, and then evaluated through multiple classifier configurations under controlled conditions.

The first stage of analysis consisted of dataset inspection and cleaning. The original dataset was examined to identify the available text and label columns, class distribution, empty statements, and very short statements. Empty statements were removed before modelling because they did not contain usable linguistic information. Very short but non-empty statements were retained, since short expressions may still contain relevant lexical or emotional cues in short-text classification.

The second stage consisted of text preprocessing and feature construction. After empty-statement removal, the text data were normalized and converted into two main forms of representation. The lexical branch used TF-IDF-based features, including standard word-level representations and a combined word-and-character configuration. The semantic branch used sentence embeddings generated with the pretrained all-MiniLM-L6-v2 model. These representations formed the analytical basis for comparing lexical and semantic modelling strategies.

The third stage consisted of supervised classification analysis. Seven candidate configurations were evaluated, combining the selected feature representations with Logistic Regression, Multinomial Naive Bayes, Random Forest, and Linear SVM, depending on the representation branch. This stage allowed the study to compare not only the general effectiveness of lexical versus semantic features, but also the interaction between representation type and classifier behaviour.

Model comparison was performed using stratified five-fold cross-validation. Stratification was used because the classification task involved seven unevenly

distributed categories, and preserving class proportions across folds helped produce more reliable performance estimates. Five-fold cross-validation was selected to reduce dependence on a single train-test split while keeping the computation feasible in the Google Colab environment.

The primary model-selection metric was Macro F1-score. Accuracy and Weighted F1-score were also reported as supporting metrics. Accuracy measures the proportion of correctly classified instances among all instances. In general terms, it can be expressed as:

$$Accuracy = \frac{\text{Number of correct predictions}}{\text{Total number of predictions}}$$

Using the confusion matrix notation, Accuracy can also be expressed as:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

In this formula, TP refers to true positives, TN to true negatives, FP to false positives, and FN to false negatives. Accuracy provides a general view of overall classification performance, but it can be misleading in imbalanced multiclass settings because it may hide weak performance on smaller or more difficult categories.

Precision, recall, and F1-score were used to evaluate class-level performance. Precision measures how many instances predicted as a given class were actually correct:

$$Precision = \frac{TP}{TP + FP}$$

Recall measures how many actual instances of a given class were correctly identified:

$$Recall = \frac{TP}{TP + FN}$$

The F1-score combines precision and recall through the harmonic mean:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

In multiclass classification, precision, recall, and F1-score are calculated at class level and then averaged depending on the selected averaging strategy. Macro F1 was obtained by calculating the F1-score separately for each class and then averaging the class-level scores equally across all classes. This was important because the study involved an

imbalanced multiclass dataset, and the objective was not only to perform well on the most frequent categories, but also to evaluate whether the model behaved reasonably across the full label set.

Weighted F1 was also reported because it averages class-level F1-scores while taking class support into account. This provides a complementary metric that reflects the influence of class frequency. However, Macro F1 remained the main selection criterion because it gives equal importance to each category and is therefore more appropriate for identifying weaknesses in smaller or more difficult classes.

After cross-validation, the best-performing configuration was selected according to mean Macro F1-score and evaluated on a final stratified hold-out test set. This final evaluation provided an independent estimate of performance on unseen data. The hold-out evaluation reported Accuracy, Macro F1-score, Weighted F1-score, and a full classification report including precision, recall, F1-score, and support for each category.

The final stage of analysis focused on error inspection and class-level behaviour. This was carried out through the classification report and the confusion matrix, which made it possible to examine how performance was distributed across categories and where misclassification persisted. These outputs were important because they moved the analysis beyond a single summary score and allowed a more careful interpretation of which categories were easier or harder to distinguish.

Collectively, these techniques supported a controlled comparative analysis of short mental-health-related text classification. The analytical strategy was therefore not limited to identifying the highest-performing model, but also aimed to clarify how representation choice affected model behaviour, performance balance across classes, and the transparency of the evaluation process.

3.7. Ethical Considerations

This study does not involve direct interaction with human participants, clinical intervention, or collection of personally identifiable information by the researcher. The analysis is conducted on a pre-existing labelled dataset used as secondary research material. For this reason, the ethical focus of the study is not on participant recruitment or informed consent procedures administered by the researcher, but on the responsible use, interpretation, and reporting of sensitive text data.

Because the dataset contains language related to mental-health concerns, the study treats the task as ethically sensitive even though it is computational and non-clinical in scope. The research does not attempt to diagnose individuals, infer medical conditions beyond the dataset labels, or make clinical claims about real persons. Instead, the models are used only to classify text statements into predefined dataset categories within a controlled experimental setting. This distinction is important in order to avoid overstating what the system can legitimately conclude [12], [21].

A second ethical consideration concerns data quality and responsible preprocessing. During the initial inspection, empty statements were identified and removed before modelling because they did not contain usable linguistic information for text classification. Very short but non-empty statements were retained because short expressions may still contain meaningful lexical or emotional cues. This decision was made to improve the quality of the modelling dataset without applying an overly aggressive filtering strategy that could remove valid short-text signals.

A third ethical consideration concerns privacy and contextual sensitivity. Since the study relies on an existing dataset, the researcher does not control the original conditions under which the texts were created, collected, or labelled. For that reason, the thesis avoids presenting the source texts as direct evidence of verified mental health status and does not frame the results as clinically authoritative. The findings are interpreted only as dataset-based computational outcomes [12], [21].

A further ethical issue involves transparency and responsible reporting. In a domain related to mental health, it is not sufficient to report performance scores without acknowledging uncertainty, class-level errors, and methodological limitations. The study therefore emphasizes controlled evaluation, explicit reporting of metrics, and cautious interpretation of misclassification patterns. This helps reduce the risk of presenting automated predictions as more reliable or generalizable than the data and experimental design actually support [12]–[14].

Finally, the study recognizes that machine learning models trained on secondary text data may reflect biases originating from dataset composition, label assumptions, or platform-specific language patterns. These risks cannot be fully removed within the scope of the present research, but they are acknowledged as an important limitation. For this reason, the thesis maintains a non-clinical framing throughout and treats the results as

methodological findings rather than as tools for real-world diagnosis, screening, or intervention [12], [21].

3.8. Ensuring Credibility and Validity

Credibility and validity in this study are supported mainly through the design and execution of the computational experiment. Since the research is based on supervised machine learning rather than on participant-based instruments, trustworthiness depends on the consistency of the analytical process, the fairness of model comparison, and the transparency with which preprocessing decisions, evaluation metrics, and results are reported.

One important measure was the use of a controlled comparison setting. All candidate configurations were evaluated on the same cleaned dataset, under the same general preprocessing logic, and within the same evaluation framework. This made the comparison more credible because differences in performance could be interpreted in relation to the representation strategy and classifier choice rather than to unrelated procedural variation.

A second measure concerned dataset inspection and cleaning before modelling. The original dataset was examined for class distribution, empty statements, and very short statements. Empty statements were removed because they did not contain usable linguistic information for text classification, while very short but non-empty statements were retained because they could still contain meaningful lexical or emotional cues. This decision improved the quality of the modelling dataset while preserving relevant short-text information.

A third measure was the use of stratified five-fold cross-validation during model comparison, followed by a final stratified hold-out evaluation for the selected model. This two-stage design reduced dependence on a single train-test split and provided a more stable basis for judging comparative performance. Stratification was especially important because the task was multiclass and the class distribution was uneven.

A fourth measure concerned the choice of evaluation criteria. The study did not rely on Accuracy alone, since overall accuracy can hide weaker performance in less frequent or more difficult classes. For that reason, Macro F1-score was treated as the main model-selection metric, while Accuracy and Weighted F1-score were reported as supporting

measures. In addition, the classification report and confusion matrix were used to examine class-level behaviour and remaining misclassification patterns.

Credibility was also supported by transparent output generation. The notebook exported the main tables and figures used in the thesis, including the class distribution table, cross-validation results, final hold-out metrics, classification report, class-level metrics, and confusion matrix. This helped maintain consistency between the computational workflow and the written thesis, reducing the risk of manually transcribing results incorrectly.

Reproducibility was also supported by exporting intermediate and final outputs from the notebook. The cross-validation table, hold-out metrics, classification report, confusion matrix, class-level metrics, and generated figures were saved as separate files. This reduced the risk of inconsistencies between the computational workflow and the thesis document. It also made the reported results easier to verify, since the values presented in the findings chapter could be traced back to specific notebook outputs.

The importance of transparent reporting is especially relevant in mental-health-related NLP. Previous work has shown that studies in this area may suffer from incomplete reporting, variable dataset quality, and limited clarity around methodological decisions [12]. More broadly, concerns about interpretability and responsible model evaluation also support the need to report not only performance scores, but also class-level behaviour, errors, and limitations [13], [14].

The study also included post hoc inspection of lexical model features and potential artefacts. This inspection was not used as clinical interpretation, but as a methodological check of what kinds of textual signals influenced the selected lexical model. The presence of some platform- or URL-related artefacts was acknowledged as a limitation, since such features may reflect dataset-specific structure rather than purely meaningful mental-health-related language. This is consistent with prior concerns that dataset composition, platform effects, and labelling assumptions can influence mental-health-related text classification results [12], [20].

At the same time, the study does not claim strong external validity in the sense of broad real-world generalization. The findings are tied to a specific secondary dataset, a defined label structure, selected representation strategies, and a non-clinical short-text classification setting. For this reason, validity in this thesis is understood mainly as the strength of the internal comparative design and the transparency of the evaluation process,

rather than as evidence that the same results would automatically transfer to all mental-health-related text environments.

3.9. Summary of Methodology

This chapter presented the methodological structure of the study. The research was framed as a quantitative, comparative, and non-clinical machine learning study based on a secondary labelled dataset of short mental-health-related text statements. The methodology was organized around a controlled comparison between lexical and semantic text representations under the same general experimental conditions.

The chapter described the dataset preparation process, including the inspection of the original corpus, removal of empty statements, preservation of short but non-empty statements, and construction of the final modelling dataset. After empty-statement removal, the dataset used for modelling contained 52,681 records distributed across seven predefined categories. This preparation step was important because it improved the quality of the textual input while preserving the short-text nature of the classification task.

The chapter also explained the computational workflow used in the study. The analysis was implemented in Python within Google Colab using standard NLP and machine learning libraries. Text data were preprocessed and represented through two alternative feature strategies: TF-IDF-based lexical representations and sentence embeddings generated with a pretrained transformer-based sentence embedding model. Seven candidate supervised learning configurations were then evaluated in order to compare representation strategies and classifier behaviour.

Model comparison was performed using stratified five-fold cross-validation, with Macro F1-score used as the main selection metric and Accuracy and Weighted F1-score reported as supporting metrics. The best-performing configuration was then evaluated on a final stratified hold-out test set. Class-level behaviour was examined through the classification report and confusion matrix, allowing the study to analyse not only total performance but also category-specific strengths, weaknesses, and misclassification patterns.

Ethical and validity-related considerations were also addressed. The study was explicitly framed as a computational text classification task rather than a clinical diagnostic system. The methodology emphasized responsible handling of sensitive text data, transparent reporting of preprocessing and evaluation decisions, and cautious interpretation of model

outputs. These methodological decisions provide the basis for the findings presented in the next chapter.

4. FINDINGS

4.1. Key Results

This chapter presents the findings obtained from the experimental comparison of the selected supervised learning configurations. The results are reported after empty-statement removal, using the final modelling dataset of 52,681 short mental-health-related text statements distributed across seven predefined categories.

The model comparison was conducted through stratified five-fold cross-validation using Macro F1-score as the main selection metric. Accuracy and Weighted F1-score were also reported as supporting metrics. Across the evaluated configurations, performance differed according to both the representation strategy and the classifier used.

The strongest global performance was obtained by the TF-IDF word-and-character + Linear SVM configuration. This model achieved the highest mean Macro F1-score during cross-validation and was therefore selected as the winning configuration for final hold-out evaluation. The result indicates that, in this dataset and experimental setting, the strongest lexical configuration outperformed the evaluated sentence-embedding-based alternatives.

In the final stratified hold-out evaluation, the selected model achieved an Accuracy of 0.7665, a Macro F1-score of 0.7269, and a Weighted F1-score of 0.7638. These results show that the model performed reasonably well at the aggregate level, while the difference between Macro F1 and Weighted F1 also indicates that performance was not equally distributed across all categories. This is important because the dataset was imbalanced, with Normal, Depression, and Suicidal accounting for most of the observations.

At the class level, the selected model showed stronger performance for some categories than for others. The Normal category achieved the highest F1-score, followed by Bipolar, Anxiety, and Depression. In contrast, Stress, Suicidal, and Personality disorder remained more difficult to classify. This uneven behaviour shows that acceptable overall performance did not translate into uniform class-level reliability.

The confusion matrix further confirmed that some errors were concentrated between specific categories. In particular, substantial confusion remained between Depression and Suicidal, suggesting overlap in the linguistic patterns associated with these labels. This

finding supports the need to interpret the model not only through global metrics, but also through class-level performance and error analysis.

In general terms, the key finding of the study is that a strong lexical representation based on TF-IDF word-and-character features combined with Linear SVM produced the best performance among the evaluated configurations. However, the findings also show that multiclass mental-health-related short-text classification remains uneven across categories and should be interpreted cautiously within the non-clinical, dataset-specific scope of the study.

4.2. Data Representation

The findings are presented through tables and visual outputs in order to make both the comparative model results and the class-level outcomes easier to inspect. In line with the evaluation design of the study, the results are reported in two stages. First, the stratified five-fold cross-validation comparison is presented as the model-selection stage. Second, the final stratified hold-out evaluation of the selected model is reported. This distinction is important because cross-validation was used to compare candidate configurations, while the hold-out test set was used to evaluate the selected model on unseen data.

Table V presents the stratified five-fold cross-validation results for the seven evaluated model configurations. The models are ranked by mean Macro F1-score, which was used as the main model-selection criterion. Macro F1-score was prioritized because the task involves seven categories and the class distribution is uneven. Unlike Accuracy, Macro F1 gives equal importance to each category and is therefore more informative when class-level performance matters.

Table V. Stratified five-fold cross-validation results ranked by mean Macro F1-score

Model	Accuracy	Macro F1	Weighted F1
TF-IDF word + character + Linear SVM	0.7699 $\pm$ 0.0040	0.7312 $\pm$ 0.0088	0.7669 $\pm$ 0.0039
TF-IDF + Logistic Regression	0.7616 $\pm$ 0.0045	0.7083 $\pm$ 0.0081	0.7561 $\pm$ 0.0045
Embeddings + Linear SVM	0.7485 $\pm$ 0.0021	0.6782 $\pm$ 0.0058	0.7424 $\pm$ 0.0021
Embeddings + Logistic Regression	0.7469 $\pm$ 0.0031	0.6779 $\pm$ 0.0074	0.7420 $\pm$ 0.0031
TF-IDF + Random Forest	0.7236 $\pm$ 0.0035	0.6409 $\pm$ 0.0059	0.7119 $\pm$ 0.0034
Embeddings + Random Forest	0.7100 $\pm$ 0.0017	0.6101 $\pm$ 0.0061	0.6925 $\pm$ 0.0013
TF-IDF + Naive Bayes	0.6831 $\pm$ 0.0024	0.5517 $\pm$ 0.0070	0.6691 $\pm$ 0.0025

The cross-validation results show that the strongest configuration was TF-IDF word + character + Linear SVM, with a mean Macro F1-score of 0.7312. This configuration also achieved the highest mean Accuracy (0.7699) and the highest mean Weighted F1-score (0.7669) among the evaluated models. The second-best configuration was TF-IDF with Logistic Regression, while the best semantic model was Embeddings + Linear SVM. This indicates that, under the selected experimental setting, the lexical configurations outperformed the sentence-embedding-based alternatives.

Fig. 4 presents the cross-validation Macro F1-score for each evaluated model. This visual comparison makes the ranking clearer and highlights the performance gap between the strongest lexical configuration and the embedding-based models.

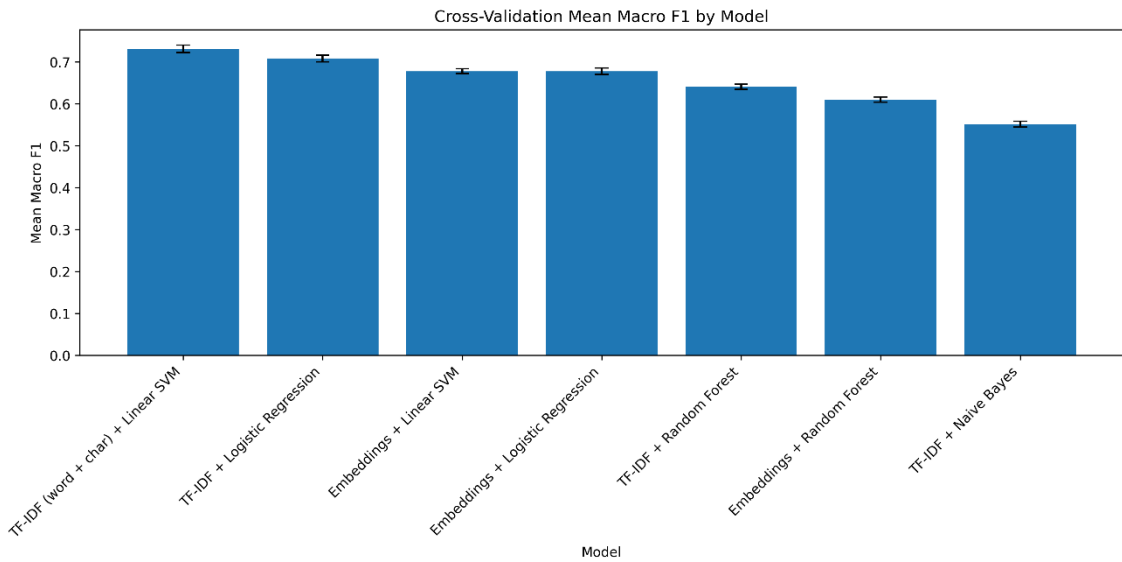


Fig. 4. Cross-validation Macro F1-score by model.

As shown in Fig. 4, the TF-IDF word-and-character representation combined with Linear SVM produced the highest Macro F1-score. The figure also shows that the two embedding-based linear models performed similarly to each other, but remained below the strongest lexical alternatives. This supports the central finding that the semantic representation branch did not provide a performance advantage over the strongest TF-IDF configuration in this experiment.

After cross-validation, the winning configuration was evaluated on a final stratified hold-out test set. Table VI reports the final hold-out performance of the selected model. This evaluation provides an independent estimate of performance on unseen data after the model-selection stage.

Table VI. Final stratified hold-out performance of the selected model

Model	Accuracy	Macro F1	Weighted F1
TF-IDF word + character + Linear SVM	0.7665	0.7269	0.7638

The final hold-out results remained consistent with the cross-validation stage. The selected model achieved an Accuracy of 0.7665, a Macro F1-score of 0.7269, and a Weighted F1-score of 0.7638. The similarity between the cross-validation and hold-out results suggests that the selected model maintained stable performance when evaluated on unseen data. However, the gap between Macro F1 and Weighted F1 also indicates that performance was not fully balanced across all categories.

Table VII presents the class-level precision, recall, F1-score, and support for the selected model on the final hold-out test set. These results make it possible to examine how performance varied across categories rather than relying only on global metrics.

Table VII. Class-level precision, recall, F1-score, and support for the selected model

Category	Precision	Recall	F1-score	Support
Anxiety	0.8025	0.7461	0.7733	768
Bipolar	0.8196	0.7536	0.7854	556
Depression	0.7054	0.7167	0.7110	3081
Normal	0.8815	0.9440	0.9117	3269
Personality disorder	0.7471	0.6047	0.6684	215
Stress	0.6483	0.5242	0.5797	517
Suicidal	0.6654	0.6523	0.6588	2131

The class-level results show that the model performed best on the Normal category, with an F1-score of 0.9117. Bipolar and Anxiety also achieved relatively strong F1-scores of 0.7854 and 0.7733, respectively. In contrast, Stress, Suicidal, and Personality disorder showed weaker class-level performance. This confirms that the model's overall performance was not evenly distributed across all categories.

Fig. 5 visualizes the class-level F1-scores of the selected model.

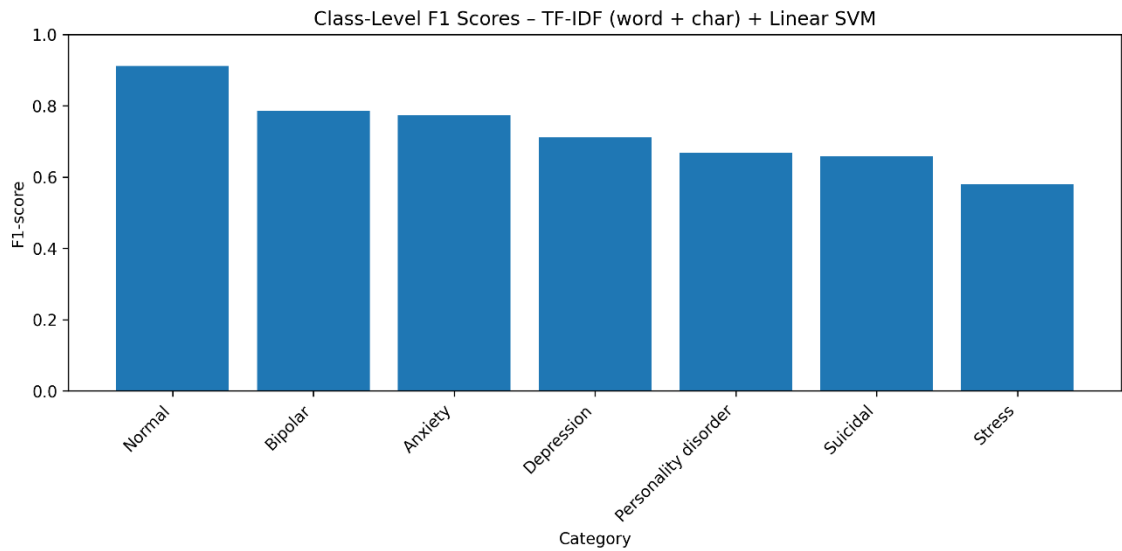


Fig. 5. Class-level F1-scores for the selected model.

As shown in Fig. 5, the class-level pattern is visually uneven. Normal is clearly separated from the remaining categories, while Stress appears as the weakest class. This pattern suggests that some labels were more linguistically distinguishable than others in the final hold-out evaluation.

Finally, Fig. 6 presents the confusion matrix of the selected model on the final stratified hold-out test set. The confusion matrix complements the tabular results by showing how predictions were distributed across the seven categories and where the most frequent misclassifications occurred.

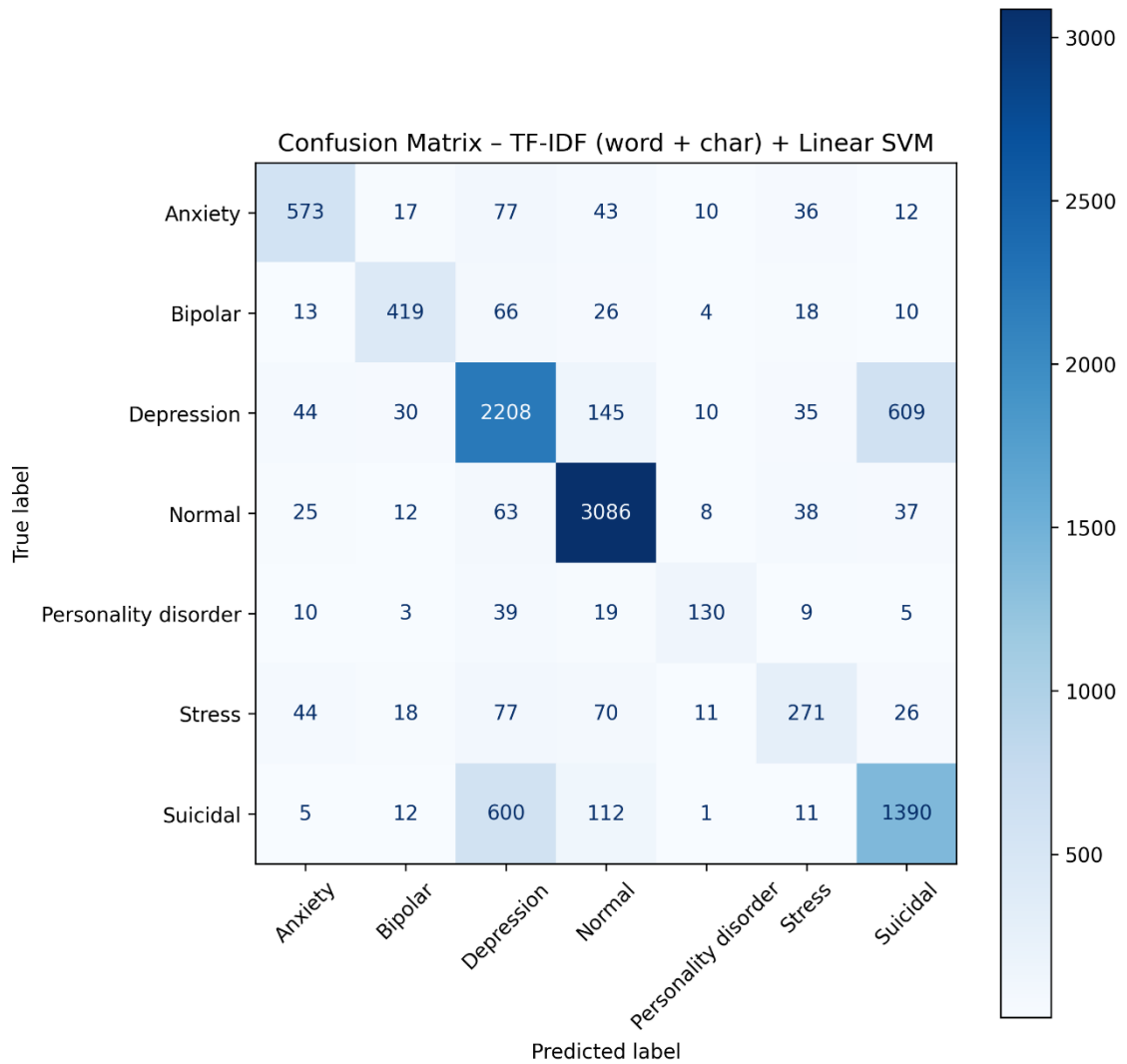


Fig. 6. Confusion matrix of the selected model on the final hold-out test set.

As shown in Fig. 6, many predictions were concentrated correctly along the diagonal, especially for Normal, Depression, and Suicidal, which were also among the largest categories. However, the matrix also reveals substantial confusion between Depression and Suicidal. In particular, several Suicidal statements were predicted as Depression, and several Depression statements were predicted as Suicidal. This suggests that these two categories share overlapping linguistic signals in the dataset. Additional confusion was also observed between Stress and Normal, and between Anxiety and Depression, indicating that short mental-health-related statements can contain ambiguous or overlapping expressions.

On the whole, the data representation shows that the TF-IDF word-and-character Linear SVM model achieved the strongest performance among the evaluated configurations and remained stable in the final hold-out evaluation. At the same time, the class-level table

and confusion matrix show that performance varied substantially across categories. Therefore, the findings should be interpreted not only through global metrics, but also through class-specific behaviour and error patterns.

4.3. Unexpected Results

One unexpected result of the study was that the strongest global performance did not come from the semantic embedding-based models, but from the lexical configuration combining TF-IDF word and character features with Linear SVM. Given the increasing use of sentence embeddings and transformer-based representations in NLP, it could reasonably have been expected that the semantic branch would achieve the strongest performance. However, this did not occur in the present experiment.

The cross-validation results showed that the TF-IDF word-and-character Linear SVM configuration achieved the highest mean Macro F1-score among all evaluated models. The best semantic configuration, Embeddings + Linear SVM, performed competitively but remained below the strongest lexical configuration. This suggests that, for this dataset, explicit lexical and subword-level patterns carried enough discriminative information to support stronger classification performance than the tested sentence-level embeddings.

This result should not be interpreted as evidence that lexical methods are generally superior to semantic representations. Rather, it indicates that representation effectiveness depends on the dataset, text length, label structure, and classifier used. In short-text classification, the available context is limited, and category-relevant information may be concentrated in specific words, expressions, punctuation patterns, or character sequences. Under these conditions, a strong lexical model can remain highly competitive.

A second unexpected result was the uneven class-level behaviour of the selected model. Although the final hold-out results showed acceptable global performance, the classification report revealed clear differences between categories. Normal achieved the strongest class-level result, while Stress, Suicidal, and Personality disorder remained more difficult to classify. This shows that the global metrics did not fully capture the variation in category-level performance.

A third unexpected result was the persistence of confusion between Depression and Suicidal. These categories are conceptually distinct in the dataset, but the confusion matrix showed substantial bidirectional misclassification between them. This suggests that the language used in some statements labelled as Depression and Suicidal may

overlap strongly, especially when expressions of hopelessness, sadness, emotional exhaustion, or distress appear without enough additional context for the model to distinguish the intended label reliably.

These findings are unexpected because they challenge two simple assumptions. The first is that semantically richer representations will automatically outperform lexical alternatives. The second is that a strong overall score implies equally strong performance across all categories. In this study, neither assumption was fully supported. The results instead suggest that short mental-health-related text classification remains highly dependent on dataset-specific lexical cues, label overlap, and class-level ambiguity.

For this reason, the unexpected results strengthen rather than weaken the methodological value of the study. They show why controlled comparisons are necessary and why evaluation should include both global metrics and class-level error analysis. The findings also support the non-clinical framing of the thesis: the model can identify statistical patterns in labelled text, but it should not be interpreted as making reliable clinical distinctions between mental health conditions.

4.4. Addressing the Research Questions

The findings address the main research question of the thesis by showing how lexical and semantic text representations performed under the same supervised multiclass classification framework. The central question was whether semantic sentence embeddings would provide better classification performance than lexical TF-IDF-based representations for short mental-health-related text statements.

The results showed that the strongest performance was achieved by the lexical configuration combining TF-IDF word-and-character features with Linear SVM. This model obtained the highest mean Macro F1-score during stratified five-fold cross-validation and was therefore selected for the final hold-out evaluation. The finding indicates that, in this dataset and under the selected experimental conditions, the lexical representation strategy was more effective than the tested sentence-embedding-based alternatives.

The results also address the question of how different classifiers behaved when combined with different representation strategies. Linear models performed strongly in both branches, but their effectiveness was higher in the lexical branch. TF-IDF with Logistic Regression achieved the second-best overall result, while Embeddings with Linear SVM

was the strongest semantic configuration. Random Forest performed less strongly in both representation branches, suggesting that it was not the most suitable classifier for this high-dimensional or embedding-based short-text classification task.

The findings further address the question of class-level performance. The selected model did not perform equally across all seven categories. Normal was the strongest category, while Stress, Suicidal, and Personality disorder remained more difficult to classify. This indicates that model performance was affected not only by representation strategy, but also by category-specific linguistic separability, class imbalance, and overlap between labels.

The confusion matrix also helped answer whether the selected model could clearly distinguish all categories. The results showed that some categories were frequently confused, especially Depression and Suicidal. This suggests that short statements assigned to these labels may share similar expressions of distress, hopelessness, sadness, or emotional exhaustion. Therefore, even the best-performing configuration did not fully resolve the ambiguity between conceptually related mental-health-related categories.

In general, the research questions were addressed by showing that the strongest lexical model outperformed the tested semantic alternatives, but also by demonstrating that the classification task remained uneven and sensitive to class-level ambiguity. The findings support the conclusion that representation choice matters, but they also show that strong global performance does not guarantee equally reliable classification across all mental-health-related categories.

5. DISCUSSION

5.1. Interpreting Results

The main finding of this study is that the strongest performance was achieved by the lexical configuration combining TF-IDF word-and-character features with Linear SVM. This result is important because the study was designed to compare whether lexical or semantic representations would be more effective for classifying short mental-health-related text statements under the same supervised learning framework. The result shows that, in this dataset and under the selected experimental conditions, the strongest lexical model outperformed the tested sentence-embedding-based alternatives.

This finding suggests that the short statements in the dataset contained explicit lexical and subword-level cues that were highly useful for classification. In mental-health-related short texts, relevant information may be concentrated in specific words, repeated expressions, short phrases, punctuation patterns, or character-level fragments. A combined word-and-character TF-IDF representation can capture these signals directly, while a sentence embedding model compresses the full statement into a dense vector. In this task, the direct representation of surface-level textual cues appears to have been more effective than the broader semantic approximation provided by the tested embeddings.

The result should not be interpreted as evidence that lexical representations are universally superior to semantic representations. A stronger interpretation is that representation effectiveness depends on the specific characteristics of the dataset, including text length, label structure, class distribution, and the type of linguistic signals available in the statements. Since the texts in this study are short, the amount of context available for semantic encoding is limited. Under these conditions, a simpler but well-tuned lexical representation may preserve discriminative information that is diluted or compressed in dense sentence embeddings.

The performance of Linear SVM is also consistent with the nature of high-dimensional text classification. TF-IDF representations often create sparse feature spaces with many word or character n-gram dimensions. Linear SVM can be effective in this setting because it can separate classes using weighted combinations of many sparse lexical features. This helps explain why the TF-IDF word-and-character Linear SVM configuration achieved the strongest performance among the candidate models.

At the same time, the findings show that the task remained difficult at class level. The selected model performed strongly for Normal and reasonably well for categories such as Bipolar, Anxiety, and Depression, but it showed weaker performance for Stress, Suicidal, and Personality disorder. This indicates that global metrics alone are not sufficient for interpreting the results. A model can achieve acceptable overall performance while still showing uneven reliability across categories.

The confusion matrix further showed that Depression and Suicidal were frequently confused. This does not necessarily mean that the model failed in a simple technical sense. Rather, it suggests that some statements assigned to these two labels may share similar linguistic expressions of distress, sadness, hopelessness, or emotional exhaustion. Because the unit of analysis is a short statement rather than a longer clinical history or conversation, the model may not have enough contextual information to distinguish between related labels when the wording overlaps.

Ultimately, the findings support the central argument of the thesis: model performance in short mental-health-related text classification depends not only on model complexity, but also on how well the representation strategy matches the structure of the input text. In this study, the strongest result came from a transparent lexical configuration rather than from a semantically richer embedding-based approach. This reinforces the need to evaluate representation strategies empirically instead of assuming that more complex or more modern representations will automatically perform better.

5.2. Relating to Literature

The findings of this study both align with and extend previous research on mental-health-related text classification. Prior studies have shown that user-generated text can contain linguistic signals associated with mental-health-related states, including depression, anxiety, suicidal ideation, and other forms of psychological distress [9]–[12], [20]. The present study supports this broader view by showing that supervised machine learning models were able to classify short labelled statements with reasonable overall performance. However, the results also show that classification quality depended strongly on the representation strategy, classifier choice, and category being predicted.

The strongest result obtained by the TF-IDF word-and-character Linear SVM configuration is consistent with previous work showing that lexical approaches remain highly competitive in text classification. Joachims emphasized that text classification

often involves high-dimensional sparse feature spaces in which many individual terms can carry useful discriminative information [6]. This is directly relevant to the present study because TF-IDF word and character features allowed the model to represent a large number of explicit lexical and subword patterns. In this setting, Linear SVM was able to use these sparse features effectively.

The result is also consistent with previous mental-health-related NLP studies that have reported strong performance from lexical and n-gram-based models. Pirina and Çöltekin found that Linear SVM with word and character n-grams performed strongly for depression identification on Reddit [20]. Similarly, Dessì et al. reported that a TF-IDF and SVM configuration outperformed a more complex embedding-based alternative in a health-related text classification setting [15]. The present study extends this line of evidence by showing a similar pattern in a seven-category short-text classification task, where the strongest lexical model outperformed the tested sentence-embedding configurations.

At the same time, the findings should not be interpreted as a general rejection of semantic representations. Sentence embeddings and transformer-based representations remain important in NLP because they are designed to capture semantic similarity beyond exact word overlap [8], [17]. Recent systematic evidence also indicates that embedding-based representations can perform strongly in text-based depression estimation when studies use standard labels such as validated scales or clinical diagnoses [24]. Therefore, the present result should be interpreted more narrowly: in this specific dataset, with short dataset-labelled statements and seven predefined categories, the tested embedding-based configurations did not outperform the strongest lexical baseline.

This distinction is important because Zhang et al. [24] focused on depression estimation using standard labels, whereas the present thesis addresses multiclass classification across seven dataset-defined categories. The tasks are related but not equivalent. Depression estimation with clinical or scale-based labels differs from classifying short statements into categories such as Normal, Anxiety, Bipolar, Stress, Personality disorder, Suicidal, and Depression. For this reason, the stronger performance of embeddings reported in some depression-focused studies does not invalidate the present finding. Instead, it shows that representation effectiveness is task-dependent and must be evaluated under the specific label structure, text length, and dataset conditions of each study.

The class-level results also relate to previous concerns about methodological caution in mental-health-related NLP. Chancellor and De Choudhury argued that mental-health prediction from social media and digital traces requires careful interpretation, transparent reporting, and awareness of ethical risks [12]. The present study supports this caution because the selected model did not perform equally across all categories. The strongest performance was observed for Normal, while Stress, Suicidal, and Personality disorder remained more difficult to classify. This shows that global metrics alone can hide uneven class-level behaviour, especially in imbalanced and conceptually overlapping classification tasks.

The confusion between Depression and Suicidal also reflects challenges discussed in prior work. Mental-health-related language is often ambiguous, emotionally overlapping, and context-dependent [12]–[14]. In short statements, expressions of sadness, hopelessness, exhaustion, or distress may not provide enough context for a model to distinguish whether the intended dataset label is Depression or Suicidal. This helps explain why the confusion matrix showed substantial bidirectional errors between these two categories. The issue is therefore not only technical, but also related to the difficulty of mapping brief natural language expressions onto discrete mental-health-related labels.

The post hoc inspection of lexical features further supports the relevance of transparent modelling. Many influential features appeared linguistically plausible, but some artefact-like features related to URLs or platform-specific tokens were also detected. This observation is consistent with concerns that dataset composition, platform effects, and labelling assumptions can influence mental-health-related text classification results [12], [20]. It also reinforces the need to treat model outputs as dataset-specific computational findings rather than as direct evidence of mental health status.

In broader terms, the findings are consistent with literature showing that mental-health-related text can be modelled computationally, but they also challenge the assumption that more semantically complex representations will always produce better results. The study contributes to the literature by providing a controlled comparison of lexical and semantic representations on the same short-text multiclass dataset. In doing so, it supports a more cautious conclusion: semantic representations are valuable, but strong lexical baselines remain necessary, especially when texts are short, labels overlap, and interpretability matters.

5.3. Study Limitations

This study has several limitations that should be considered when interpreting the results. First, the analysis is based on a secondary labelled dataset, which means that the researcher did not control the original process through which the texts were created, collected, or annotated. As a result, the quality of the labels, the assumptions behind category assignment, and the global composition of the corpus may all have influenced the observed performance. This is especially relevant in mental-health-related NLP, where dataset construction and labelling assumptions can strongly affect model behaviour and interpretation [12], [20].

Second, the findings are limited to a specific non-clinical short-text classification setting. Although the dataset contains language related to mental-health concerns, it does not provide clinically validated diagnostic evidence. The categories used in the study are treated as dataset-defined labels rather than verified medical diagnoses. Therefore, the results should not be interpreted as directly generalizable to real patients, clinical populations, screening procedures, or diagnostic practice.

Third, the dataset remained imbalanced after empty-statement removal. Normal, Depression, and Suicidal accounted for most of the observations, while Personality disorder remained the smallest category. This imbalance may have affected model learning and class-level performance. Although Macro F1-score and stratified splitting were used to reduce the risk of overemphasizing majority classes, the smaller categories still had less training data available and remained more difficult to classify.

Fourth, the task itself involved linguistic overlap between categories. The confusion matrix showed substantial confusion between Depression and Suicidal, and weaker performance was also observed for Stress, Suicidal, and Personality disorder. These errors may reflect not only model limitations, but also ambiguity in the dataset labels and overlap in the way distress is expressed in short texts. Short statements often provide limited context, which makes it difficult to distinguish between related mental-health-related categories based only on surface language.

Fifth, the semantic representation branch used a pretrained sentence embedding model without task-specific fine-tuning. The all-MiniLM-L6-v2 model provided a general-purpose sentence representation, but it was not adapted to the specific label structure or linguistic characteristics of the dataset. For this reason, the comparison should be

interpreted as a comparison between a task-sensitive lexical pipeline and a general pretrained embedding-based approach, rather than as a definitive evaluation of all semantic or transformer-based methods.

Sixth, the scope of model comparison was intentionally limited. The study evaluated seven supervised learning configurations, which was sufficient for a controlled comparison between selected lexical and semantic representations. However, it did not include fine-tuned transformer classifiers, larger embedding models, deep neural architectures, hierarchical classification methods, or multi-label formulations. These alternatives may produce different results under other experimental conditions.

A further limitation emerged from the post hoc inspection of lexical features. Although many influential features were linguistically plausible, some URL- or platform-related artefacts were also detected. This suggests that part of the learned signal may reflect dataset-specific structure rather than purely category-relevant language. The presence of such artefacts does not invalidate the comparative findings, but it limits how confidently the learned lexical patterns can be interpreted as meaningful mental-health-related cues.

A final limitation concerns the distinction between internal and external validity. The internal comparison in this thesis is relatively strong because all candidate configurations were evaluated under the same dataset, preprocessing logic, and evaluation framework. However, this does not imply strong external validity. The results may change if the same models are applied to a different dataset, a different label structure, longer texts, clinically validated labels, or a domain-adapted embedding model. For this reason, the findings should be interpreted as evidence about the behaviour of the selected methods under the present experimental conditions, not as a general rule about lexical and semantic representations in all mental-health-related NLP tasks.

5.4. Practical Impacts

The practical value of this study lies mainly in its methodological implications for short-text classification tasks involving mental-health-related language. The results show that a relatively simple and computationally efficient lexical approach can remain highly competitive, even in a research area where semantically richer representations may initially appear more attractive. This is relevant for researchers and practitioners who need models that are not only effective, but also easier to implement, compare, and interpret.

One practical implication is that strong lexical baselines should not be dismissed too quickly. In short and noisy text settings, TF-IDF-based representations combined with suitable classifiers may offer a useful balance between performance, computational cost, and transparency. This is especially relevant in small-scale research projects, educational settings, or resource-constrained environments, where larger semantic models may be less practical to train, evaluate, or justify methodologically.

A second implication concerns model selection in sensitive domains. The findings suggest that choosing a more complex representation does not automatically lead to better classification results. For applied work, this means that methodological transparency and fair comparison are just as important as the final score itself. Researchers working with mental-health-related language data should therefore evaluate simpler alternatives seriously before assuming that denser semantic approaches will perform better.

A third implication relates to class-level monitoring. The uneven results across categories show that acceptable global performance can still hide weaker behaviour in specific classes. In practical research settings, any model built for similar classification tasks should therefore be assessed not only through global metrics, but also through class-level outputs and error patterns. Otherwise, important weaknesses may remain hidden behind a single summary score.

The findings also have implications for responsible interpretation. Since the study is based on dataset-defined labels rather than clinical diagnoses, the results should not be used for real-world diagnosis, screening, or intervention. Instead, the study can inform how future computational systems in this area may be designed and evaluated more carefully. In particular, the results support the use of transparent baselines, stratified validation, Macro F1-score, classification reports, and confusion matrices as part of responsible model assessment.

Finally, the study highlights the importance of checking dataset quality and possible artefacts before interpreting model outputs. The removal of empty statements improved the quality of the modelling dataset, while the post hoc inspection of lexical features showed that some learned patterns may reflect platform- or URL-related artefacts. This reinforces the need for careful preprocessing, transparent reporting, and cautious interpretation when working with sensitive short-text datasets.

The main practical contribution of the study is therefore not a deployable mental-health system, but a transparent evaluation framework for comparing lexical and semantic representations in short mental-health-related text classification. This framework may support future research aimed at developing more reliable, interpretable, and ethically cautious NLP models in sensitive domains.

6. CONCLUSION

6.1. Summary of Results

This study compared lexical and semantic text representations for the supervised multiclass classification of short mental-health-related statements. The final modelling dataset contained 52,681 records after removing empty statements that did not contain usable linguistic information. Seven candidate configurations were evaluated under the same general experimental framework using stratified five-fold cross-validation, followed by a final stratified hold-out evaluation of the selected model.

The strongest global performance was obtained by the TF-IDF word-and-character Linear SVM configuration. This model achieved the highest mean Macro F1-score during cross-validation and was therefore selected as the best-performing configuration. In the final hold-out evaluation, it achieved an Accuracy of 0.7665, a Macro F1-score of 0.7269, and a Weighted F1-score of 0.7638.

These results show that, in the specific dataset and experimental setting used in this thesis, the strongest lexical representation outperformed the tested sentence-embedding-based alternatives. This finding supports the view that semantically denser representations do not automatically produce better performance in short-text classification tasks. Instead, the effectiveness of a representation strategy depends on the nature of the dataset, the length of the text, the label structure, and the classifier used.

The class-level results showed that performance was uneven across the seven categories. Normal achieved the strongest class-level result, while Anxiety, Bipolar, and Depression also showed relatively solid performance. Stress, Suicidal, and Personality disorder remained more difficult to classify. The confusion matrix also showed substantial overlap between Depression and Suicidal, suggesting that some categories shared similar linguistic expressions of distress in short statements.

The results therefore answer the research questions by showing that supervised machine learning can classify short mental-health-related statements with reasonable performance, but also that model behaviour remains strongly dependent on representation choice and class-level separability. The findings support the methodological value of comparing lexical and semantic representations under the same evaluation framework, rather than assuming that one representation type is automatically superior.

6.2. Connection to Literature

The results of this thesis are consistent with previous research showing that mental-health-related language can contain computationally useful signals for supervised classification. Prior studies based on social media and user-generated text have shown that linguistic patterns can be associated with depression, anxiety, suicidal ideation, and related forms of psychological distress [9]–[12], [20]. The present study supports this general view, while also showing that the usefulness of these signals depends strongly on representation strategy, classifier choice, and label structure.

The finding that TF-IDF word-and-character features combined with Linear SVM produced the strongest performance is also consistent with earlier text classification research emphasizing the strength of linear models in high-dimensional sparse text spaces [6]. It further aligns with studies showing that lexical and n-gram-based approaches can remain competitive in mental-health-related or health-related text classification tasks [15], [20]. In this thesis, the lexical configuration appears to have captured explicit word-level and character-level patterns that were highly informative for the seven-category classification task.

At the same time, the findings should be interpreted in relation to recent evidence supporting semantic and embedding-based methods. Zhang et al. [24] found that embedding-based features and deep architectures showed strong performance in text-based depression estimation studies using standard labels. This does not invalidate the present result, because the task examined in this thesis is different. Zhang et al. focused on depression estimation using standard labels such as validated scales or clinical diagnoses, whereas this thesis examined multiclass classification of short statements into seven dataset-defined categories.

Therefore, the present study contributes a cautious and task-specific conclusion. Semantic representations remain important in mental-health-related NLP, but they do not automatically outperform strong lexical baselines in every setting. In this dataset, under the selected preprocessing, modelling, and evaluation framework, the strongest lexical configuration achieved better performance than the tested sentence-embedding configurations. This comparison reflects a task-sensitive lexical pipeline against a pretrained semantic model, all-MiniLM-L6-v2, used without task-specific fine-tuning.

This supports the broader methodological point that representation choice should be evaluated empirically rather than assumed in advance.

The thesis also reinforces previous concerns about transparency and careful reporting in mental-health-related NLP [12]–[14]. The class-level results and confusion matrix showed that global metrics were not enough to understand model behaviour. Some categories were easier to distinguish, while others showed substantial overlap, especially Depression and Suicidal. This finding supports the need to report not only overall scores, but also class-level performance, error patterns, dataset limitations, and non-clinical interpretation.

The main connection to the literature is therefore twofold. First, the study supports the established view that mental-health-related text can be modelled computationally. Second, it challenges the simplified assumption that newer semantic representations are always preferable to lexical approaches. The contribution of this thesis lies in showing, under a controlled short-text multiclass setting, that strong lexical baselines remain necessary for fair and methodologically transparent comparison.

6.3. Future Research Directions

Several directions for future research emerge from the findings and limitations of this study. A first extension would be to evaluate a broader range of semantic approaches, including larger sentence embedding models and fine-tuned transformer-based classifiers. In the present thesis, the semantic branch relied on a pretrained sentence embedding model used without task-specific fine-tuning. Future work could therefore examine whether semantic representations adapted directly to the dataset and label structure produce stronger performance in short multiclass mental-health-related text classification.

A second direction would be to investigate the effect of preprocessing decisions more systematically. The present study removed empty statements before modelling while retaining very short but non-empty statements. Future research could compare alternative preprocessing strategies, such as stricter artefact removal, URL filtering, emoji normalization, spelling correction, or different stopword and lemmatization settings. This would help determine how much classification performance depends on preprocessing choices rather than representation strategy alone.

A third direction would be to examine category overlap more explicitly. The confusion observed between Depression and Suicidal suggests that some labels may share similar linguistic expressions of distress in short texts. Future studies could explore hierarchical classification, multi-label classification, or alternative annotation schemes that better reflect overlap between mental-health-related categories. This may be especially relevant when short statements do not contain enough context to support clean separation between related labels.

A fourth direction would be to include stronger interpretability methods. The present study included post hoc inspection of lexical features and potential artefacts, but it did not apply more advanced explanation techniques such as SHAP, LIME, or systematic error annotation. Future work could use these methods to examine which textual cues influence predictions and whether those cues are linguistically meaningful, dataset-specific, or artefact-driven.

A fifth direction would be to compare machine learning predictions with expert human judgement under controlled annotation conditions. This would help clarify whether model errors reflect technical limitations, ambiguous labels, or genuine disagreement in how short mental-health-related statements are interpreted. Such work could also examine whether computational models are more useful as decision-support tools than as stand-alone classifiers in sensitive domains.

Finally, future research could move beyond isolated short statements and examine longer contextual sequences, such as user timelines, conversations, or longitudinal text patterns. More context may help models distinguish between related categories and reduce ambiguity in cases where short statements alone are insufficient. However, such extensions would require more carefully controlled datasets, stronger privacy safeguards, and evaluation designs that remain transparent and ethically cautious.

6.4. Recommendations

Based on the findings of this study, several recommendations can be made for future work in short mental-health-related text classification. First, researchers should include strong lexical baselines when evaluating newer semantic or embedding-based approaches. The results of this thesis show that TF-IDF word-and-character features combined with Linear SVM can remain highly competitive in short and noisy multiclass text settings. For this reason, lexical methods should not be treated as outdated without empirical comparison.

Second, future studies should use controlled comparison designs whenever possible. When representation strategy, classifier choice, preprocessing, validation method, and evaluation metrics are changed at the same time, it becomes difficult to determine which factor is responsible for performance differences. Comparative studies should therefore isolate methodological variables more carefully and evaluate competing approaches under the same dataset and evaluation framework.

Third, researchers should report class-level results in addition to global metrics. The present study showed that acceptable overall performance can still hide weaker behaviour in specific categories. Precision, recall, F1-score, support, and confusion matrix analysis should therefore be included when evaluating multiclass mental-health-related text classification models. This is especially important when the dataset is imbalanced or when some categories are conceptually related.

Fourth, future work should treat mental-health-related labels with caution unless they are supported by clinical validation or expert annotation. In studies based on secondary or dataset-defined labels, model outputs should be described as computational classification results rather than diagnostic conclusions. This distinction is necessary to avoid overstating the meaning of predictions in a sensitive domain.

Fifth, dataset quality should be inspected and reported transparently before modelling. Empty statements, extremely short inputs, platform artefacts, URL fragments, duplicates, or unclear labels can influence model behaviour. Future studies should document how such cases are handled and should examine whether model performance is partly driven by dataset-specific artefacts rather than meaningful linguistic cues.

Finally, future research should evaluate semantic models under stronger task-adapted conditions before drawing broad conclusions about lexical versus semantic representations. Since the semantic branch in this thesis used a pretrained sentence embedding model without task-specific fine-tuning, later studies should test whether fine-tuned transformer models or domain-adapted embeddings improve performance under the same short-text multiclass setting. This would help clarify whether the advantage observed for the lexical model in this thesis reflects the dataset structure, the representation strategy, or the absence of task-specific adaptation in the semantic branch.

The main recommendation from this thesis is therefore methodological: representation choice should be tested, not assumed. In short mental-health-related text classification,

strong lexical baselines, transparent reporting, class-level evaluation, and cautious non-clinical interpretation should be treated as minimum requirements for responsible model comparison.

REFERENCES

- [1] World Health Organization, “Depressive disorder (depression),” WHO, Aug. 29, 2025. [Online]. Available: <https://www.who.int/news-room/fact-sheets/detail/depression>. [Accessed: Apr. 14, 2026].
- [2] World Health Organization, Depression and Other Common Mental Disorders: Global Health Estimates. Geneva, Switzerland: WHO, 2017.
- [3] J. W. Pennebaker, R. L. Boyd, K. Jordan, and K. Blackburn, The Development and Psychometric Properties of LIWC2015. Austin, TX, USA: University of Texas at Austin, 2015.
- [4] M. Tausczik and J. W. Pennebaker, “The psychological meaning of words: LIWC and computerized text analysis methods,” *Journal of Language and Social Psychology*, vol. 29, no. 1, pp. 24–54, 2010.
- [5] S. Bird, E. Klein, and E. Loper, *Natural Language Processing with Python*. Sebastopol, CA, USA: O’Reilly Media, 2009.
- [6] T. Joachims, “Text categorization with support vector machines: Learning with many relevant features,” in *Proc. European Conference on Machine Learning (ECML)*, 1998.
- [7] A. Vaswani et al., “Attention Is All You Need,” in *Proc. Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
- [8] N. Reimers and I. Gurevych, “Sentence-BERT: Sentence embeddings using Siamese BERT-networks,” in *Proc. 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, Hong Kong, China, 2019, pp. 3982–3992, doi: 10.18653/v1/D19-1410.
- [9] M. De Choudhury, S. Counts, and E. Horvitz, “Predicting postpartum changes in emotion and behavior via social media,” in *Proc. SIGCHI Conference on Human Factors in Computing Systems (CHI)*, Paris, France, 2013, pp. 3267–3276, doi: 10.1145/2470654.2466447.
- [10] G. Coppersmith, M. Dredze, and C. Harman, “Quantifying mental health signals in Twitter,” in *Proc. Workshop on Computational Linguistics and Clinical Psychology*:

From Linguistic Signal to Clinical Reality, Baltimore, MD, USA, 2014, pp. 51–60, doi: 10.3115/v1/W14-3207.

- [11] D. E. Losada and F. Crestani, “A test collection for research on depression and language use,” in *Experimental IR Meets Multilinguality, Multimodality, and Interaction: 7th International Conference of the CLEF Association, CLEF 2016, Évora, Portugal, Sep. 5–8, 2016*, pp. 28–39, doi: 10.1007/978-3-319-44564-9_3.
- [12] S. Chancellor and M. De Choudhury, “Methods in predictive techniques for mental health status on social media: A critical review,” *npj Digital Medicine*, vol. 3, no. 1, Art. no. 43, 2020, doi: 10.1038/s41746-020-0233-7.
- [13] Z. C. Lipton, “The mythos of model interpretability,” *Queue*, vol. 16, no. 3, pp. 30–57, 2018.
- [14] F. Doshi-Velez and B. Kim, “Towards a rigorous science of interpretable machine learning,” *arXiv preprint arXiv:1702.08608*, 2017.
- [15] D. Dessì, R. Helaoui, V. Kumar, D. R. Recupero, and D. Riboni, “TF-IDF vs Word Embeddings for Morbidity Identification in Clinical Notes: An Initial Study,” in *Proc. 1st Workshop on Smart Personal Health Interfaces (SmartPhil)*, Cagliari, Italy, 2020, pp. 1–12.
- [16] J. W. Pennebaker, R. Booth, and M. Francis, *Linguistic Inquiry and Word Count: LIWC 2001*. Mahwah, NJ, USA: Lawrence Erlbaum Associates, 2001.
- [17] J. Camacho-Collados and M. T. Pilehvar, “From word to sense embeddings: A survey on vector representations of meaning,” *Journal of Artificial Intelligence Research*, vol. 63, pp. 743–788, 2018, doi: 10.1613/jair.1.11259.
- [18] L. Breiman, “Random Forests,” *Machine Learning*, vol. 45, pp. 5–32, 2001, doi: 10.1023/A:1010933404324.
- [19] G. Coppersmith, C. Harman, and M. Dredze, “Measuring post-traumatic stress disorder in Twitter,” in *Proc. International AAAI Conference on Web and Social Media (ICWSM)*, 2014.
- [20] I. Pirina and Ç. Çöltekin, “Identifying depression on Reddit: The effect of training data,” in *Proc. 2018 EMNLP Workshop SMM4H: The 3rd Social Media Mining for*

Health Applications Workshop & Shared Task, Brussels, Belgium, 2018, pp. 9–12, doi: 10.18653/v1/W18-5903.

- [21] B. G. Teferra, A. Rueda, H. Pang, R. Valenzano, R. Samavi, S. Krishnan, and V. Bhat, “Screening for Depression Using Natural Language Processing: Literature Review,” *Interactive Journal of Medical Research*, vol. 13, p. e55067, 2024, doi: 10.2196/55067.
- [22] S. Hameed, M. Nauman, N. Akhtar, M. A. B. Fayyaz, and R. Nawaz, “Explainable AI-driven depression detection from social media using natural language processing and black box machine learning models,” *Frontiers in Artificial Intelligence*, vol. 8, 2025, doi: 10.3389/frai.2025.1627078.
- [23] T. Tseriotou, J. Chim, A. Klein, A. Shamir, G. Dvir, I. Ali, C. Kennedy, G. S. Kohli, A. Hills, A. Zirikly, D. Atzil-Slonim, and M. Liakata, “Overview of the CLPsych 2025 Shared Task: Capturing Mental Health Dynamics from Social Media Timelines,” in *Proceedings of the 10th Workshop on Computational Linguistics and Clinical Psychology (CLPsych 2025)*, Albuquerque, NM, USA, 2025, pp. 193–217, doi: 10.18653/v1/2025.clpsych-1.16.
- [24] S. Zhang, C. Zhang, and J. Zhang, “Text-based depression estimation using machine learning with standard labels: Systematic review and meta-analysis,” *Journal of Medical Internet Research*, vol. 28, p. e82686, 2026, doi: 10.2196/82686.

ANNEX A. REPRODUCIBILITY MATERIALS

This annex summarizes the computational materials generated during the implementation of the study. The main experiment was conducted in a Google Colab notebook, and the main outputs were exported as CSV and PNG files in order to preserve consistency between the computational workflow and the written thesis. These materials support transparency and make it possible to trace the results reported in the Findings chapter back to the notebook outputs.

The notebook included the main stages of the experiment: dataset loading, empty-statement inspection, removal of empty records, text preprocessing, feature construction, stratified five-fold cross-validation, final hold-out evaluation, classification report generation, confusion matrix generation, lexical feature inspection, artefact checking, and export of tables and figures. The exported files were used to reduce the risk of manually transcribing results incorrectly.

The purpose of this annex is not to reproduce the full notebook code, but to document the main computational files generated during the experiment. These outputs provide supporting evidence for the tables, figures, and results presented in the thesis.

The exported files were stored together with the implementation notebook in the project repository to support result traceability and future verification. Some exported files were renamed with shorter descriptive names for clarity while preserving their original content.

The final implementation notebook and exported outputs are available in the project repository listed below. The repository includes the notebook file, exported CSV files, generated PNG figures, dataset source information, and supplementary materials used to support the results reported in the thesis.

Repository: <https://github.com/SergioVillarreal18/mental-health-text-classification-thesis>.

Table A1. Exported computational outputs used in the thesis

Output	File name	Purpose
Dataset source information	data_source/README.md	Provides information about the original Kaggle dataset source and explains that the dataset itself is not redistributed in the repository.
Candidate model configurations	candidate_model_configurations.csv	Documents the seven candidate model configurations evaluated in the experiment.
Class distribution after empty-statement removal	class_distribution_after_empty_removal.csv	Stores the class distribution used in Table IV and Fig. 3.
Class-level metrics	class_level_metrics.csv	Stores class-level precision, recall, F1-score, and support.
Classification report	classification_report.csv	Stores the full classification report for the selected model.
Confusion matrix	confusion_matrix.csv	Stores the confusion matrix values used for class-level error analysis.
Cross-validation results	cross_validation_results.csv	Stores the stratified five-fold cross-validation results reported in Table V.
Empty-statement impact by class	empty_statement_impact_by_class.csv	Documents how empty-statement removal affected each category.
Final hold-out results	final_holdout_results.csv	Stores the final performance metrics reported in Table VI.
Suspicious word-level artefacts	suspicious_word_level_artefacts.csv	Stores potentially artefact-like tokens identified during post hoc inspection.
Top SVM features	top_svm_features.csv	Stores the top lexical features inspected after training the selected model.
Class distribution figure	figure_class_distribution_after_empty_removal.png	Visualizes the class distribution after empty-statement removal.
Class-level F1-score figure	figure_class_level_f1_scores.png	Visualizes class-level F1-scores for the selected model.

Confusion matrix figure	figure_confusion_matrix_selected_model.png	Visualizes the final confusion matrix of the selected model.
Cross-validation Accuracy figure	figure_cv_accuracy_by_model.png	Visualizes cross-validation Accuracy by model.
Cross-validation Macro F1 figure	figure_cv_macro_f1_by_model.png	Visualizes cross-validation Macro F1-score by model.
Cross-validation Weighted F1 figure	figure_cv_weighted_f1_by_model.png	Visualizes cross-validation Weighted F1-score by model.
Final hold-out metrics figure	figure_final_holdout_metrics.png	Visualizes the final hold-out performance metrics of the selected model.
Notebook file	thesis_implementation_notebook.ipynb	Contains the computational workflow used to produce the exported results.
Project repository	GitHub repository link	Provides access to the implementation notebook, exported outputs, generated figures, and supplementary materials used to support reproducibility and result traceability.

The exported outputs strengthen the reproducibility of the study because they provide a direct link between the notebook implementation and the written thesis. The numerical outputs support the reported cross-validation results, final hold-out metrics, class-level performance table, and confusion matrix. The PNG outputs support the visual figures included in the Findings chapter.

The notebook file associated with these outputs was saved as `thesis_implementation_notebook.ipynb`. This file contains the computational workflow used to produce the results after the methodological correction involving empty-statement removal. The exported files were stored together with the notebook to support result traceability and future verification.

ANNEX B. POST HOC LEXICAL FEATURE INSPECTION

This annex provides supplementary methodological detail about the post hoc lexical feature inspection conducted after selecting the TF-IDF word-and-character Linear SVM model. While Annex A lists the exported computational outputs used in the thesis, the present annex focuses specifically on the interpretation and methodological role of two supplementary inspection outputs: `top_svm_features.csv` and `suspicious_word_level_artefacts.csv`. The purpose of this inspection was not to interpret the model clinically, but to examine whether the strongest lexical model relied on plausible textual cues, dataset-specific artefacts, or a mixture of both.

Since the selected model used TF-IDF word and character features, the learned feature weights could be inspected more directly than in the sentence-embedding branch. This supports methodological transparency because lexical features correspond to observable textual units. However, the presence of a highly weighted word or character pattern should not be interpreted as clinical evidence. It only indicates that the feature contributed to class separation within the selected dataset and model configuration.

The inspection helped clarify why the lexical configuration may have performed strongly in this experiment. Short mental-health-related statements may contain explicit words, fragments, punctuation patterns, or character-level signals that are useful for supervised classification. A word-and-character TF-IDF representation can preserve these cues directly, while dense sentence embeddings compress the full statement into a general semantic vector.

At the same time, the inspection also identified some potentially artefact-like tokens among the lexical features. These included URL- or platform-related fragments that may reflect dataset-specific structure rather than meaningful mental-health-related language. This finding supports the limitation discussed in the thesis: some learned lexical signals may be influenced by the composition and formatting of the dataset.

For this reason, the feature inspection was treated as a methodological check rather than as an explanation of mental health status. The inspected features were used to assess transparency and possible artefacts in the selected model, not to make diagnostic claims about individuals or categories. This distinction is important because the thesis remains a non-clinical computational classification study.

Table B1. Methodological role of the lexical inspection outputs

Inspection element	Purpose
Top SVM lexical features	To examine which word-level or character-level patterns received high positive weights for each class.
Word-level artefact check	To identify whether URL-, platform-, or formatting-related tokens appeared among influential features.
Class-specific feature patterns	To assess whether some learned features appeared linguistically plausible for dataset-defined categories.
Artefact identification	To document possible dataset-specific signals that should not be interpreted as clinically meaningful.
Methodological interpretation	To support transparency and cautious reporting of the selected lexical model.

The post hoc inspection showed that lexical feature analysis can provide useful methodological insight, but it also has clear limits. Feature weights help identify which textual patterns contributed to classification within the trained model, but they do not explain psychological states, clinical conditions, or individual experiences. Therefore, the inspected features should be understood only as model-specific indicators within the selected dataset.

This annex reinforces one of the central methodological points of the thesis: interpretability is useful only when it is framed carefully. Lexical models are more inspectable than dense embedding-based models, but inspection does not automatically produce clinical explanation. In sensitive domains such as mental-health-related text classification, transparent feature analysis must be combined with cautious interpretation, artefact checking, and explicit non-clinical framing.